

A review on the effect of micro- and nano-plastics pollution on the emergence of antimicrobial resistance

Vivek Kumar Gaur^{a,b,c,1}, Ranjna Sirohi^{d,1}, Mohd Ishfaq Bhat^e, Krishna Gautam^a, Poonam Sharma^f, Janmejai Kumar Srivastava^b, Ashok Pandey^{a,g,h,*}

^a Centre for Energy and Environmental Sustainability, Lucknow, 226 029, India

^b Amity Institute of Biotechnology, Amity University Uttar Pradesh, Lucknow Campus, Lucknow, India

^c School of Energy and Chemical Engineering, Ulsan National Institute for Science and Technology, Ulsan, 44919, Republic of Korea

^d Department of Food Technology, School of Health Sciences and Technology, University of Petroleum and Energy Studies, Dehradun, 248 007, Uttarakhand, India

^e Department of Post-Harvest Process and Food Engineering, Govind Ballabh Pant University of Agriculture and Technology, Pantnagar, 263145, India

^f Department of Bioengineering, Integral University, Lucknow, India

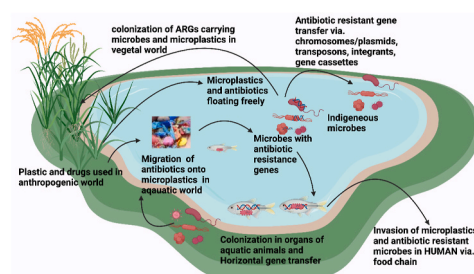
^g Sustainability Cluster, School of Engineering, University of Petroleum and Energy Studies, Dehradun, 248 007, India

^h Centre for Innovation and Translational Research, CSIR-Indian Institute of Toxicology Research, Lucknow, 226 001, India

HIGHLIGHTS

- Microplastics and antimicrobial resistance has become a global menace.
- Microplastics serve as breeding ground for AMR microorganisms.
- They also interact with other persistent pollutant in the environment.
- Combating strategies should be focused considering both the pollutants together.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Antimicrobial resistance
Antibiotics
Microplastics
Plastic pollution

ABSTRACT

The recent upsurge in the studies on micro/nano plastics and antimicrobial resistance genes has proven their deleterious effects on the environmental and human health. Till-date, there is a scarcity of studies on the interactions of these two factors and their combined influence. The interaction of microplastics has led to the formation of new plastics namely plastiglomerates, pyroplastics, and anthroplastics. It has long been ignored that the occurrence of microplastics has become a breeding ground for the emergence of antimicrobial resistance genes. Evidently microplastics are also associated with the occurrence of other pollutants such as polycyclic aromatic hydrocarbons and pesticides. The increased use of antibiotics (after Covid breakout) has further elevated the detrimental effects on human health. Therefore, this study highlights the relation of microplastics with antibiotic resistance generation. The factors such as uncontrolled use of antibiotics and negligent plastic consumption have been evaluated. Furthermore, the future research prospective was provided that can be helpful in correctly identifying the seriousness of the environmental occurrence of these pollutants.

* Corresponding author. Centre for Innovation and Translational Research, CSIR-Indian Institute of Toxicology Research, Lucknow, 226 029, India.

E-mail addresses: ashokpandey1956@gmail.com, ashok.pandey1@iitr.res.in (A. Pandey).

¹ Authors contributed equally.

1. Introduction

Plastics, in the contemporary world, are a class of ineluctable materials being versatile in nature supplemented by its processing easiness, durability, ability to be tailored for specific needs, aesthetic appeal, resistance to corrosive materials, availability in abundance, and exceptional thermal properties (Romeo, 2020). Owing to this, plastic production worldwide has increased from a mere 1.5 MMT in 1950 to 368 MMT in 2019, with a slight decrease to 367 MMT in 2020 due to Covid-19 pandemic and its effect on the industry (Tiseo, 2021). More than a quarter of this production is done by China, while about 57 MMT of the production in 2019 was from Europe alone (Tiseo, 2021). The adverse effects of AMR and microplastics has been extensively reported in recent years (Dong et al., 2021; Gaur et al., 2022; He et al., 2021). Globally by 2050 the annual death toll due to AMR could rise to 10 million (de Kraker et al., 2016; Pires et al., 2017). The indiscriminate, inappropriate, and excessive use of therapeutically relevant antibiotics in the healthcare, veterinary, and agriculture sectors, which may reach human population through a variety of routes. The lack of understanding is the primary cause of growing AMR in the current environment (Cui et al., 2020; Samreen et al., 2021; Wang et al., 2022). These aspects lead to the introduction of multidrug-resistant bacterial outbreaks in the population by increasing genetic selection pressure. The most common pathogens in this group are *Salmonella* spp. and *Campylobacter* spp. In addition, microorganisms obtained from humans and animals have exhibited similar resistance mechanisms.

Heavy metals, microplastics (MPs) and nanoplastics (NPs), antibiotics, and other environmental contaminants have a collective influence on the persistence, development, migration, and lifecycle of antibiotic-resistant bacteria (Bartkova et al., 2021; Cui et al., 2021). Among above said environmental pollutants, such as MPs (1 μm –5 mm) and NPs ($\leq 1 \mu\text{m}$) are known to produce plastisphere once particles interacted with microorganisms because plastic particles give a surface for microbes to thrive on (Frias and Nash, 2019; Gigault et al., 2018). Micro (nano)plastics (MNPs) are synthetic plastic particles, fibers, and films that have been found in both aquatic and terrestrial environments (Kumar et al., 2021; Lusher, 2015; Mohammad Mirsoleimani Azizi et al., 2021). These pieces of plastic can absorb organic and inorganic micro-nutrients from the environment, attracting bacteria, viral particles, and other microbes to their surfaces (Frère et al., 2018). Consequently, microplastic have become breeding ground for drug-resistant bacteria (Dong et al., 2021). Due to the limits of present analytical techniques, nanoplastic have undergone lesser research than microplastic; however, researchers have projected that in future, nanoplastic in mass might be 10^{14} times greater than currently recorded MP concentrations (Besseling et al., 2019; Nguyen et al., 2019). MNPs can also assist microorganisms in enduring environmental challenges and improving microbial diffusivity by providing relatively stable environments (Oberbeckmann et al., 2015). This functionality of MNPs might help to improve materials flow and energy transmission in the food chain, and it could also be one of the reasons why bacteria stick to their surfaces. Very few studies have identified the interaction between micro/nanoplastics and environmental factors. However, since these compounds are strictly water insoluble, therefore, they do not significantly affect the DO or pH of water. Yet, it has an indirect relation as the global increase in the temperature affects the DO, and other parameters, therefore, it may cause an increase the water toxicity of the micro/nanoplastics (Gao et al., 2022).

Bacterial resistance evolves as a result of natural selection. Resistance develops genetically via mutation or by acquiring additional resistance genes through genetic exchange processes (Blakely, 2014; López-Velandia et al., 2018). MNPs can promote gene transfer between attached biofilm populations or between biofilm communities and their surroundings (Arias-Andres et al., 2018) (Fig. 1). The growing resistance of harmful bacteria to popular antibiotics derived from human and livestock settings across the world indicates the critical need for enhanced surveillance and research to prevent AMR from spreading

further (Chaturvedi et al., 2021; Dadgostar, 2019; Essack, 2018; Larsson and Flach, 2021). According to several studies, microplastics can enrich antimicrobial resistance genes, and microorganisms from the surrounding water (S. Wang et al., 2020; Yang et al., 2019). Because of the wide diversity of bacterial genes, horizontal gene transfer in the environment can result in gene exchange across microorganism groups, and new bacteria can emerge (Shen et al., 2019). As a response, microplastics that carry antibiotic-resistance genes generate genetic pollution, traveling from the ocean circulation to the coast. It is transmitted through the food supply chain from enriched species to aquaculture. Antibiotic use in cattle and poultry has lately grown globally, and in highly polluted industrialized and developing nations, it is anticipated to increase by 67% by 2030 (Van Boeckel et al., 2015). In the downstream waterways of healthcare wastewater treatment plants, resistant genes and bacteria are frequently identified (Karkman et al., 2018).

Understanding the combined influences and mechanisms resulting from the interaction of these two factors (MPs and ARGs) is critical and necessary for improving the safety of healthy aquaculture and aquatic product quality, environmental protection, and improving the treatment of contaminants found in wastewater (Dong et al., 2021; Zhang et al., 2021). Therefore, this review was focused on addressing the pollution and negative consequences posed by MPs alone and in conjunction with other emerging pollutants with special emphasis on their interaction and association with AMR. The present AMR scenario before and after the Covid-19 epidemic is also discussed in detail. Furthermore, this study also examines existing resistance mitigation measures and significant information gaps to gain a better understanding of antimicrobial resistance propagation in the environment. These characteristics may provide new insights towards fighting AMR-related concerns, which have become more prevalent because of MPs' persistence over the world.

2. Plastics pollution

2.1. Micro- and nano-plastic

The plastic waste is increasing globally in the terrestrial, aquatic, and atmospheric environment and has become a serious environmental problem. During last few decades the annual production of plastics have increased to 8.7% (Rai et al., 2021). The fragmentation or breakdown of plastic waste into smaller particles resulted in the formation of microplastics (particle size 0.1 μm –5 mm) and nanoplastics (particle size 0.001 μm –0.1 μm) (Allouzi et al., 2021). Other sources of micro-, nano-plastic includes personal care products, rubber particles, polymer, atmospheric deposition of fragments and films and fibers from industry and households (Liu et al., 2022). The microplastics and nanoplastics that originated from the plastics waste negatively impacted the environment and human health (Rai et al., 2021). The presence of MNPs in the ecosystem induces adverse effects and acts as a carrier for several contaminants, including metals, persistent organic pollutants (POPs), by transferring toxicity to biota and pathogenic microorganisms serving as a habitat for microbes; bacteria and viruses. The exposure of micro- and nano-plastics in humans occurs by inhalation of contaminated air or via intake of microplastics contaminated food such as edible fish and other sea foods, drinking water and consumer products like toothpaste, beer, and honey. that causes potential risks to human health such as cancer, loss of reproduction, mental retardation, and other related organ problems (Prüst et al., 2020). A study was reported for MNP contamination of animals (Chicken) that enters the human food chain. In this, the chicken polluted with microplastics was studied as a subject. The chicken gizzards were eaten by the Southeast Mexican population and reported to contain MPs in gizzards with a concentration of 10.2 ± 13.8 particles (Allouzi et al., 2021; Toussaint et al., 2019). Toussaint et al. (2019) studied the contamination of microplastics in the marine species and reported that around 18%–33% of catfish collected from Brazilian estuaries contains plastic particles in their stomachs. The consumption of small fish and seafoods acts as a direct source for contamination of

microplastics in the human diet.

The European Food Safety Authority (EFSA) reported the concentration of microplastics in fish, shrimps, and bivalves, to be 1–7 particles/g, 0.75 particles/g, and, 0.2 to 4 microparticles/g respectively. Similarly, researchers also studied the presence of MPs in *M. edulis* along United Kingdom coasts. They recorded microplastics ranging from 0.7 to 2.9 items per g in all the collected samples (Toussaint et al., 2019). Furthermore, the toxicological effect of micro- and nano-plastics on benthic invertebrates such as phytoplanktons and crustaceans in the aquatic ecosystem has been evaluated (Haegerbaeumer et al., 2019). Several researchers also examined the impact of micro and nanoplastics in soil system. The microplastics affects the plant growth, physio-chemical properties of the soil, polluting the groundwater as these plastics exist in the soil and interact with potentially toxic elements (PTE) and organic pollutants in the environment, causing soil toxicity. MNPs can be of high density and low density; the former retains in soil moving into the inner layers and transferring and contaminating groundwater, plants, and food web while the later remains at the surface and transported via wind and water erosion (Allouzi et al., 2021).

There is limited literature for understanding the fate of microplastic in the environment. Since these are more persistent in the environment, many studies have identified the movement of these MNP in terrestrial and aquatic environment (Fig. 2) (Boyle and Örmeci, 2020). It was reported that in sandy soil the LDPE microplastics from <150 µm to 1 mm showed movement towards the bottom due to the burrowing caused by *Lumbricus terrestris* (Yu et al., 2019). It was reported that these microplastic showed 50 cm downward movement in 14 d. Most interestingly, the movement was not only due to the ingestion and egestion but also due to the adherence of microplastics to the earthworm exterior (Boyle and Örmeci, 2020). The behavior and fate of MNP in terrestrial and freshwater is governed by many complex interactions including physico-chemical property of soil, water infiltration and soil dwelling organisms. More detailed studies need to be performed to develop better understanding regarding the influence of different parameters on the behavior and fate of MNP in different ecosystems (Boyle and Örmeci, 2020).

2.2. New plastics in Anthropocene

Plastic has become a widely used material due to its versatility, adaptability, utility, resistance, low-cost production, and light weight since it was initially created in the early 20th century. Plastic has become an inevitable component of modern epoch where it is profoundly available as a base material for feeder bottle to saline bottle, bricks to roads and dresses to electronic gadgets like laptops. The accumulation of plastic contaminants in environment can also be witnessed from remote areas like protected natural reserves, glaciers, agro-ecosystems, terrestrial habitats, to arctic deep-ocean sediments (De-la-Torre et al., 2021; Malizia and Monmany-Garzia, 2019). The adherence of plastic traces with geological and archaeological records could be a distinctive attribute of Anthropocene (Mytum and Meek,

2021). The extraordinary durability of plastics imposes severe long-lasting consequences on Earth's geology. Plastic had performed crucial role in technological revolution while 'Great Acceleration' of population, industrialization and resources demand after World War II.

At the starting phase of the 20th century, natural materials such as polysaccharides (cellulose, gelatine), proteins (casein), and rubber were used to synthesize natural biopolymers as a man-made plastics, by the middle of 20th century, natural plastics had been absolutely replaced by petrochemical-based polymers (Porta, 2021). Synthetic polymers or plastics are covalently attached chains of repeating organic macromolecules which are available as remoldable thermoplastics (e.g. polyethylene, polystyrene, polypropylene, nylon, polyvinyl chloride, and polyethylene terephthalate) or rigid thermosets (e.g. polyurethane, melamine formaldehyde, epoxy and phenolic resins) for application in microelectronics components like circuit boards (Taffel, 2016).

As per an estimate around 8300 million metric tons (Mt) of plastic is manufactured globally till 2017. Out of 5 billion tons plastic manufactured till 2015, 6300 million Mt of plastic waste was generated, approximately 12% was incinerated, 9% of which has been recycled; and almost 79% has been dumped in landfills or environments (Malizia and Monmany-Garzia, 2019). This amount of plastic is enough to wrap the Earth in a cling wrap or plastic cover. Viewing the current trends, it is estimated that around 40 billion tons of plastic would be manufactured by 2050 leading a burden of 12,000 million Mt plastic waste as landfill or in nature, this amount is sufficient to wrap 6 layers of cling film around our planet. It is an alarming issue that majority of the global plastics produced so far, is still persistent in the environment (Malizia and Monmany-Garzia, 2019; Zalasiewicz et al., 2016).

Depending upon the size, plastics is divided broadly as macroplastics (>5 mm) and microplastics (<5 mm), both of which are causing socio-ecological problems and challenges in anthropocene. Disposal of hygiene products or cigarette filters (1 filter tip contains ~10,000 fibres) leads to accumulation of rayon microparticles in the environment (Zalasiewicz et al., 2016). A significant proportion of microplastics (38–234m⁻³ particles) were reported to be found in frozen Arctic sea and was reward as a major global sink of microparticles (Zalasiewicz et al., 2016). Upon reacting with marine conditions, plastic waste yields novel formations with varying configuration, physical and chemical characteristics, namely Plastiglomerates, pyroplastics, plasticrusts, and anthropoquinas (Table 1). Plastiglomerates are new type of rock, formed of multi-composite matrix of molten plastic, sediment or beach sand, stones, shells, basaltic lava debris and traces of organic material due to anthropogenic activities like burning, campfires (De-la-Torre et al., 2021).

Pyroplastics, a sub-type of plastiglomerates, is a floating amorphous matrix of burnt or molten plastic carrying a neutral shade of black-charcoal-grey to off-white or brown, with hues of blue, green, yellow, or pink color. It is seldom accompanied with cracks, cavities, fractures, and pits. Due to weathering in sea surface, pyroplastics attains smooth surfaces but release toxic chemical additives into marine biota (Silva et al., 2016). Hexabromocyclodecanes (HBCDs) and 1,2-dibromo-4-(1,

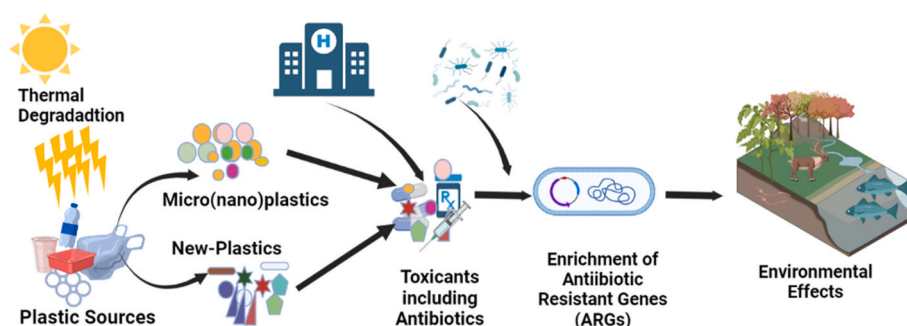


Fig. 1. Emergence of antibiotic resistance genes on microplastics.

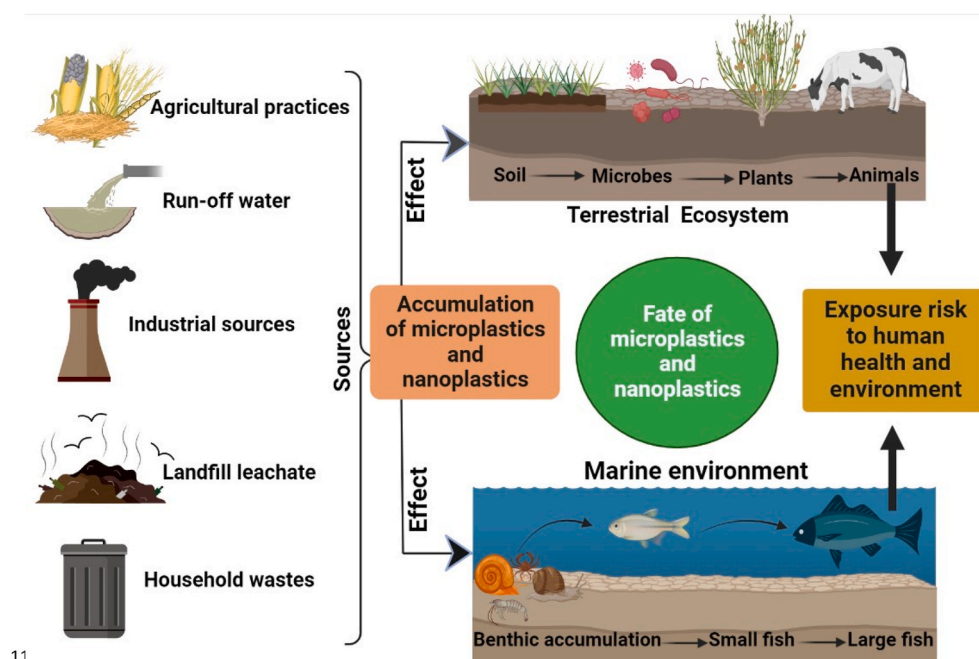


Fig. 2. Sources and fate of micro- and nano-plastic in terrestrial and aquatic environment.

Table 1

Types of new plastic, source fate and impact.

Type of plastic	Polymer type	Source	Fate	Method for characterization	Environmental impact	References
Plastiglomerate	–	Burning	Burial	–	Chemical release	Corcoran et al. (2014)
Anthropoquina	–	Litter in sedimentary rocks	Geological record	–	Aesthetics	Fernandino et al. (2020)
Pyroplastic	Polyethylene terephthalate	Burning and weathering	Persistence	ATR-FTIR	Chemical release and ARG transport	Ehlers and Ellrich (2020)
Pyroplastic	Polypropylene	Wave action	Fragmentation and Persistence	ATR-FTIR	Plastic ingestion	Turner et al. (2019)
Plasticrust	Polyethylene	Wave action	Fragmentation and Persistence	ATR-FTIR	Plastic ingestion	(De-la-Torre et al., 2021; Ehlers and Ellrich, 2020)

2-dibromoethyl)-cyclohexane (TBECH) are brominated flame retardants profoundly used as plastic pose detrimental effects on behavioural, reproductive health of biota and marine food web (Ruan et al., 2018). Bisphenol A (BPA) is a plastic-associated additives, which causes endocrine-disruption in living bodies and environment (De-la-Torre et al., 2021). Plasticrust (term coined by Gestoso et al. (2019)) are encrusted rocks like texture of plastics, formed due to crash of larger plastic items against rocks via. Coastal waves. Fernandino et al. (2020) invented the term anthropoquinas, to describe encrusted sedimentary beach rocks that contain anthropogenic materials such as plastics, glass, brick fragments, slags, wood and other organic waste. These anthropogenic artefacts are termed as “techno fossils”, and are expected to follow the footsteps of plastiglomerates to become a part of the geological cycle (De-la-Torre et al., 2021).

2.3. Current scenario of antimicrobial resistance (AMR)

The most serious human health threat of the 21st century has been identified as antimicrobial resistance (AMR), which jeopardize the control and efficient treatment of an ever-expanding variety of parasitic, bacterial, viral, and fungi against which antibiotics are no longer effective to the widely accepted medication for humans use (Prestinaci et al., 2015). The WHO (World Health Organization) published the first worldwide report on AMR surveillance in April 2014. The data was collected nationally and internationally via monitoring networks (Prestinaci et al., 2015). Antimicrobials, which include antibiotics, have

helped people live longer lives all around the world. Antimicrobial resistance (AMR) causes treatment inefficiency and persistent infections, resulting in increased risk of severe illness (Rodríguez-Baño et al., 2021). A lack of comprehensive measures to combat AMR is projected to result in millions of deaths every year worldwide, and by 2050, AMR could overtake cancer as the leading cause of death (Ukuhor, 2021). Over 670,000 illnesses and 33,000 fatalities owing to resistant bacteria were reported by the ECDC (European Centre for Disease Prevention and Control) in 2019 (ECDC, 2020), while resistant fungi and bacteria caused at least 35,900 deaths and 3 million infections in the United States alone (Center for Disease Control and Prevention, 2019).

2.4. Pre- and post-covid effect of antimicrobial resistance

The World Health Organization (WHO) proclaimed the COVID-19 (coronavirus disease 2019) epidemic, which was introduced by the variant SARS-CoV-2, an outbreak on March 11, 2020, and has redefined the world (WHO, 2020). It has a significant influence on healthcare systems, the market, environment, and financial backing worldwide. During the COVID-19 pandemic, the top priority is to treat patients and prevent healthcare workers from contracting SARS-CoV-2 while keeping healthcare systems operational (Rodríguez-Baño et al., 2021). On the other hand, the COVID-19 outbreak is expected to impact bacterial pathogen transmission and the emergence of resistance to antibiotics, i. e. AMR (Rodríguez-Baño et al., 2021). AMR, a similar cross-sectoral problem that has gotten more international attention since 2015, has

been already designated as a global public health priority for the year 2020 (Knight et al., 2021; WHO, 2015). Due to a lack of staff and the necessity to transfer clinical diagnostics platforms, consumables, monitoring, and traceability from AMR surveillance to COVID-19 diagnostics, AMR monitoring and disease surveillance to detect colonization by resistant microbes have been hampered (Rodríguez-Baño et al., 2021). Therefore, the clinical dynamics of COVID-19 disease and therapy have changed throughout the pandemic, with new research guidelines being added regularly. Steroids are indisputably the gold standard in the treatment of COVID-19 infection, particularly in hypoxemic individuals. However, there has been inappropriate use of steroids in the treatment of mild to moderate symptoms, creating a perfect environment for secondary infections. COVID-19 disease can show in various ways, with a predominance of respiratory discomfort, fever, and sore throat, or it can be asymptomatic (Saini et al., 2021). Meanwhile, virus pathogens that cause serious disease are typically associated with immunosuppression, subjecting individuals to secondary infections that can complicate the disease's course by increasing morbidity and death (Saini et al., 2021).

Globally the most significant health concern was antimicrobial resistance (AMR) before the pandemic because of excessive/misuse and improper disposal of unused antibiotics; however, with the beginnings of SARS-CoV-2, all initiatives were decided to focus on COVID-19 and its control, resulting in some compromise on antimicrobial therapy and auditing (Murray, 2020; van Hengel and Marin, 2019). Antibiotic usage was low before the Covid 19 epidemic due to fewer primary health care prescriptions (Andrews et al., 2021). However, there are fears that the COVID-19 outbreak could jeopardize recent progress in practical antibacterial usage, which prevents individuals from infections caused by *Clostridioides difficile* following antibiotic treatment and aids in the combat against AMR (Huttner et al., 2020). Therefore, in May 2020, the WHO issued interim recommendations on the clinical treatment of COVID-19 to improve antimicrobial safety awareness and guidelines of antimicrobial management (World Health Organization, 2020). It does not suggest antimicrobial treatment or prevention for mildly ill patients to intermediate COVID-19 symptoms. Still, second-line antibiotic medication for individuals with chronic COVID-19 is recommended, depending on medical decision-making and de-escalation evaluations (Andrews et al., 2021). Despite its capacity to collect resistance mechanisms and thrive in hospital surroundings, *A. baumannii* was only responsible for 1.7% of illnesses related to resistant bacteria recorded by the ECDC before the COVID-19 pandemic (ECDC, 2020). The questions about the danger of microbial alone or significant co-infections during COVID-19 have been raised since the beginning of the SARS-CoV-2 outbreak.

Furthermore, any associated infections induced by a microorganism are commonly referred to as co-infections other than SARS-CoV-2 that were present during COVID-19. In contrast, all pathogenic activities are considered secondary infections that were not present at the time of diagnosis but occurred later in the disease (Langford et al., 2020). Therefore, the use of anti-MRSA medications has grown in on-bed ICU patients with lungs-related severe difficulties due to the presence of methicillin-resistant (MRSA) and methicillin-sensitive (MSSA) bacteria with covid-19 infection, implying bacterial secondary infection (Elabadi et al., 2021). In addition, Wide-ranging antibiotic treatment was given to approximately 86.4% of all COVID-19 infected patients admitted to the ICU to ensure effective protection against bacterial bronchial co-infections (Langford et al., 2021). The most commonly found AMR strain were *Escherichia coli*, followed by *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Enterococcus faecalis*, and *Pseudomonas aeruginosa* (ECDC, 2020). *E. Coli* was shown to be the most dangerous microorganism that causes bloodstream infection (BSI) worldwide among the identified antibiotic resistance bacteria (Sawatwong et al., 2019). Consequently, critical patients' reliance on invasive mechanical ventilation, lengthy hospitalization, mortality rates, and uncertainty in the identification and prognosis of secondary bacterial infection have

led to the administration of antimicrobials on an empiric basis. According to Rawson et al. (2020) 72 percent of patients were administered antibiotics, despite the fact that only 8% of them were confirmed to have a fungal or bacterial infection.

Nowadays, excessive use of alcohol-based disinfectants and hand sanitizers, without proof of their acute and chronic harmful effects, poses significant health problems to humans, animals, and the environment (Mantlo et al., 2020). Consequently, medicinal and non-pharmaceutical chemicals have been exposed to the undiscovered world of bacterial diversity continually, at various frequencies and concentrations and AMR spread is facilitated by mutagenesis events that result in microbial genomic instability. Furthermore, horizontal drug resistance gene transfer can disseminate antimicrobial resistance genes (ARGs) between various bacterial species, aggravating the situation further (Nayak and Parshionikar, 2021). Furthermore, according to reported studies the usage of non-antibiotic medications has also been connected to the emergence of AMR (Y. Wang et al., 2020). These studies are important for comprehending antibiotic-resistant in the perspective of the SARS-COV-2 outbreak, which resulted in the widespread usage of non-pharmaceuticals such as alcohol-based sterilization chemicals that are now widely used as a technique of infection prevention (Hui et al., 2020).

Furthermore, COVID-19-related healthcare restrictions hampered access to appropriate diagnosis, medication, and control of individuals with serious infections requiring antibiotics. Therefore, awareness, and action are needed to combat the human health problems associated with complete drug-resistant, multidrug-resistant, and drug-resistant microorganisms (Lobie et al., 2021). To summarize, as a result of the global reaction to the pandemic, disinfectants and cleaning agents, as well as medicines, have been widely used, perhaps contributing to an increase in antimicrobial resistance (AMR). For decades, the effect of multidrug-resistant microorganisms on health and economics has been reported. Some governments have acknowledged the need for a coordinated effort to address AMR mitigation through expanded worldwide awareness and monitoring.

2.5. Correlation between AMR and plastics

The plastic wastes get accumulated in agriculture soils, marine and freshwater ecosystems (directly or through land based sources) causing environmental plastic pollution which is the most pressing global issue in the contemporary era. Plastics undergo degradation through chemical, thermal, photooxidation and biodegradation but at extremely slow pace (Alshehrei, 2017; Leal Filho et al., 2019; Ma et al., 2022; Sun et al., 2021a, 2021b). However, they undergo biophysical disintegration during the course of their life-cycle into macro (>5 mm), micro (100 nm-5mm) and nano-plastics (<100 nm) attributed to physical shearing, ultraviolet exposure, hydrolysis which initiates weathering of polymer chain backbone (Gerritse et al., 2020; Yang et al., 2019). Of these, only microplastics has been extensively researched. The problems of identification and characterization of nano-plastics have led to the its subjugation (Joo et al., 2021). Thus, only microplastics will be discussed from here onwards. The plastic pollution (land, air or water) on macro-scale has a widespread effect directly or indirectly on the animal and human health, ecosystem, environment, and climate due to the presence of toxic compounds like phthalates, poly-fluorinates, bisphenol A (BPA), brominated flame retardants and antimony trioxide (Alabi et al., 2019; Campanale et al., 2020). However, the scaled down micro and nano-plastics (either on the land or aquatic environment) have more severe implications due to large surface area, ease and rapidity of spreading in marine environment, ingestion by smaller invertebrates (zooplankton) thereby accumulating in the food web, direct uptake by the plants, ability to affect the ecosystem (through carbon sequestration, and biogeochemical cycling) and the more novel problem, at the supra-molecular level, of acting as an ecological niche for evolution and transfer of metallic and antimicrobial resistance genes (ARG's) (Azeem

et al., 2021; Gerritse et al., 2020; Leal Filho et al., 2019; Rai et al., 2021). It is an astonishing fact that 20 trillion particles of 1nm^3 volume are generated by a bag made of two plastic sheets of the size $50\text{ cm} \times 40\text{ cm} \times 50\text{ }\mu\text{m}$ thick (Gerritse et al., 2020). Owing to limited knowledge of antimicrobial resistance and its remedial, and supplemented by the abundance and ubiquitous presence of micro and nano-plastics, the problem of antimicrobial resistance poses an unprecedented challenge to the mankind and as such, world health organization has listed it as one of the ten global public health threats faced by humans at present.

2.6. Antimicrobial resistance

Microplastics have been found to be involved in the evolution and proliferation of AMR through a tripartite association of microplastics, microbes and antimicrobials or ARG's, enhancing the horizontal ARG's transfer and posing a threat to human health (Yang et al., 2019). Microplastics are quite ubiquitous especially in aquatic environments besides soils across the world and has become proliferative niche (called "Plastisphere") for microorganisms (Courten-Jones et al., 2017; Tamminga et al., 2018; Woodall et al., 2014; Zhang et al., 2017) besides their association with metals, co-contaminants, persistent organic pollutants (discussed in next section) and antimicrobial substances (Fred-Ahmadu et al., 2020). Microplastics and ARGs are considered emerging pollutants with great concern because of their adverse impacts on human and environmental health. This has led to the increase in the colonization of degrader microbial strains, which has become a favorable condition for ARG evolution (Liu et al., 2021). The microplastics having ARGs cause genetic pollution that flows from the ocean currents to the offshore and then is transmitted from the enriched organisms offshore to the aquaculture through the food chain. This suggests that the aquaculture environment had a significant role in the selective enrichment, transformation, and effect of microplastics and ARGs interactions. Besides, forming a new microbial niche microplastic acts as a reservoir for ARGs in the marine ecosystem that serves as a route for the transport of ARGs and other pollutants such as heavy metals and antibiotics (Li et al., 2018; Ma et al., 2020; Su et al., 2021). As of now, more than 200 subtypes of ARGs have been quantified in aquatic environments using metagenome, conventional qPCR, and high-throughput PCR (Dong et al., 2021; Liu et al., 2021). Lu and colleagues quantified four subtypes of ARGs using real-time PCR, tetracycline resistance (tetB, tetG, and tetX), sulfonamides (sul1 and sul2), macrolides (ermF and ermT), and quinolones (qnrA, qnrB, and qnrS) (Lu et al., 2019). The sulfonamide resistance genes were dominant in water, followed by tetracycline and quinolone resistance genes. The absolute abundance of a total of 10 ARGs in the water sample and on microplastics ranged from 3.24×10^5 to 7.83×10^5 copies/mL and 1.59×10^9 to 1.83×10^9 copies/g, respectively. The relative abundance of ARGs in microplastic/water ranged (5.15–111)/(0.51–2.57) copies per 16 S rRNA (Lu et al., 2019). The shotgun metagenomic analysis was performed to investigate the biofilm's microbial abundance and antibiotics resistance on biodegradable and non-biodegradable plastics staged at the Northern Gulf of Mexico (Sun et al., 2021a, 2021b). Interestingly, they did not observe any significant difference in the Shannon diversity and abundance of ARGs on the biodegradable and non-biodegradable plastics namely polyhydroxyalkanoate (PHA) and polyethylene terephthalate (PET) respectively. Furthermore, it was reported that the multidrug resistance genes were less abundant (2.05 copies per 16 S rRNA) on PHA than those on the PET (3.05). Beta diversity revealed the difference in the resistome of the PHA and PET. *Pseudomonas* was identified in both the biofilms as host for multidrug resistance genes and glycopeptides, whereas on PET sheets the *Desulfovibrio* was identified as the macrolide-lincosamide-streptogramin (MLS) primary host (Sun et al., 2021a, 2021b).

Several researchers studied the occurrence of ARGs on microplastics in different environments such as wastewater treatment plants (WWTPs), aquatic and terrestrial habitats, and air. WWTPs were

considered as the hotspot for microplastics and ARGs. The study revealed that the enrichment of ARGs on microcosm and microplastics study on WWTPs and landfill leachate primarily depends on the type of polymer, ARGs subtype, environment, and incubation period (Liu et al., 2021). It was reported that the use of triclosan, an antimicrobial compound employed during polymer synthesis, has led to the increase in environmental pollution by inciting the spread of ARGs. It was recorded that bacitracin, aminoglycoside, multidrug resistance genes, and macrolide-lincosamide-streptogramin were dominated among the 64 ARGs isolated from microplastics (Rinanda et al., 2019). Fuyi et al. studied the effects of microplastics on ARGs in estuary sediments using high throughput qPCR methods and found that microplastics changed the composition of ARGs in sediments.

In wake of the evidence of microplastic presence in soil and environment, the plastisphere can now be divided in "Aquatic Plastisphere", "Soil Plastisphere" and "Atmospheric Plastisphere". The atmospheric microplastics have evidence of origin from sprays brought about by breezes emanating from the sea. In a pilot study on the marine boundary layer of French Atlantic coast, it was found that the microplastic particles were present in both onshore (avg. 2.9 MP/m^3) and offshore (avg. 9.6 MP/m^3) (Allen et al., 2020). The AMR microbes associated microplastics, if ingested by marine organisms or zooplanktons can enter the food web and endanger the human and animal welfare. Studies have also revealed the uptake of microplastics by plants (Azeem et al., 2021). Also, it will lead to more serious need of inventing new antimicrobial agents (a quite tedious work) (Duval et al., 2019). The illustration below shows the association of microplastics with other entities (Plastisphere) and their routes of occurrence and entrance into the food web (Fig. 3). The microplastics act as platform for the development of antimicrobial through a cycle initiated by biofilm development, co-existence of antimicrobial agents and microbial colonies, development of ARG's and finally, horizontal gene transfer to other microorganisms as illustrated in Fig. 4. This phenomenon can take place in aquatic, soil or atmospheric plastisphere.

Bacterial, fungal and even viral pathogen associations with microplastic or/and AMR have been found to occur in aquatic, terrestrial and atmospheric plastispheres (Gkoutselis et al., 2021; Liu and Schauer, 2021; Zhu et al., 2021). In a study conducted on the total suspended solids in air (38 samples at three), in the largest medical centre in Latin America, it was found the 22 samples tested positive for the SARS-CoV-2 with predominant (80%) polyester microplastics (Amato-Lourenço et al., 2022). This study also showed a positive correlation between number of both microplastic particles and the SARS-CoV-2. A recent review has been done on the projected overwhelming increase in AMR due to the Covid-19 pandemic (Rizvi and Ahammad, 2022).

Of the cases mostly studied, on microplastic association with AMR, only pristine microplastics have been examined. However, studies have shown that aged microplastics have more potential for enhancing ARG transfer. Recently, a study showed the significance of UV-aged microplastics in enhanced ARG propagation through synergistic effects confounded with leaching of co-contaminants from microplastics. The study revealed that after 20 days of UV aging of polystyrene microplastics the ARG transfer efficiency respectively increased by 1.3, 4.7- and 3.5-fold for *E. coli*, plasmid pET29 and phage lambda (relative to pristine microplastic). The characterization of microplastics and associated contaminants, AMR and composition of microbial communities using various sophisticated techniques in detail (Bartkova et al., 2021).

Interestingly, it was noted that polyvinyl chloride (PVC) and polyethylene (PE) significantly changed the composition and structure of ARGs. In contrast, polyvinyl alcohol (PVA) reduced the number of ARGs present in the sediments, and the absolute abundance of ARGs increased to 4.1×10^9 , 8.1×10^9 and 2.0×10^9 copies/g for PVC, PE, and PVA microplastics, respectively (Huang et al., 2019). These findings in recent years have suggested that the microplastics not only themselves had served as a pollutant but have also become the breeding ground for other notorious toxicants. Thus, the distribution of ARGs on microplastics

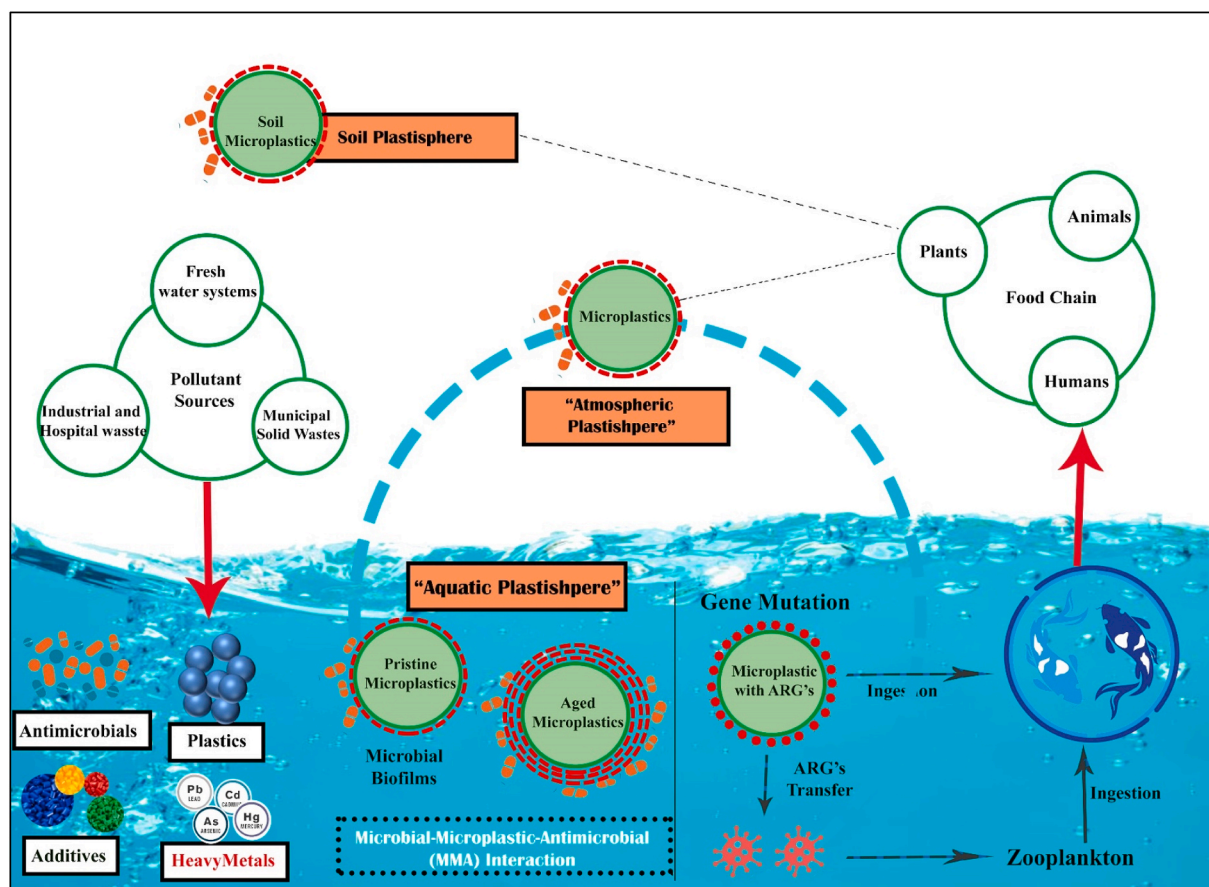


Fig. 3. Association of Microplastic-Microbes and Antimicrobials, their sources and routes to food chain.

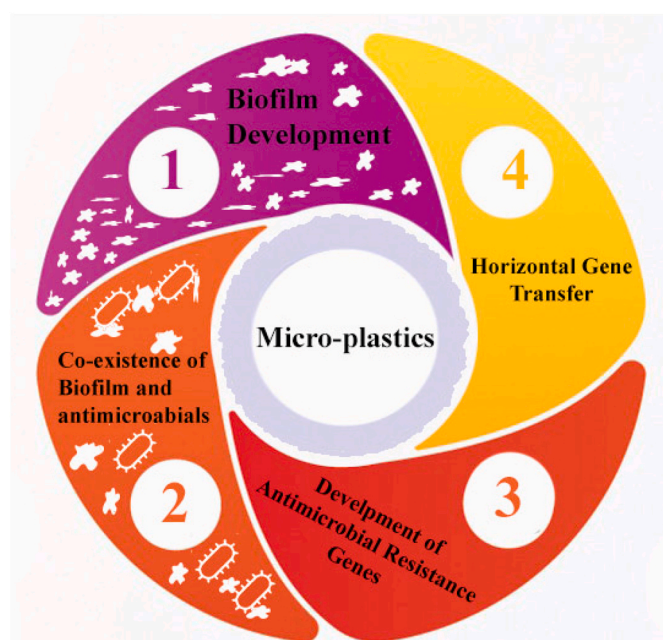


Fig. 4. Cycle of ARG development and Horizontal Gene transfer.

needs more detailed studies to decipher the ARGs enrichment process further.

2.7. Other hazardous pollutants

The intrinsic associative properties, environmental conditioning and ageing of microplastics with other toxic materials along with biotic and abiotic factors are formidable (Johansen et al., 2018; Joo et al., 2021). This is also accrued by its ubiquitous presence and the exponential increase especially in aquatic environment (Kaza et al., 2018). Microplastic associate with intrinsic co-contaminants (plastic additives like bisphenol A, nonylphenol) and extrinsic materials like metals (main metals viz. Aluminium, Iron, Manganese, and trace metals viz. Copper, Zinc, Lead, Cadmium, Silver) and persistent organic pollutants (e.g polychlorinated butyls (PCB's), bis-2-ethylhexyl phthalate (DEHP), phenanthrene (Phe), dichlorodiphenyltrichloroethane (DDT), phthalic acid ester's (PAE)) in addition to microorganisms (Ashton et al., 2010; Bakir et al., 2016; Holmes et al., 2012). The dangers pertaining to this association results from their leakage or transfer after ingestion of associated microplastics by marine organisms. The interaction of heavy metals with microplastics and their fatal effects on human and animal health have been extensively reviewed (Table 2) (Liu et al., 2021; Wang et al., 2019). Experimental studies have shown the uptake pathway of contaminants through microplastics by marine biota is of less significance but in the coming times is invariably very alarming (Bakir et al., 2016).

Table 2

Association of microplastic (virgin or biofilm associated) with metallic and organic pollutants.

S. No	Pollutant	Sample (Sediment or Microplastics)	Experimental conditions and characterizing methodology	Results (inferences)	References
Metals					
1.	^{109}Cd , ^{134}Cs , ^{65}Zn	- Virgin PE (32–75 μm) and Sediment 35–91 μm	- Exposure time (4, 24 and 72 h) - Estuarine water - γ -spectrometer counting	- 2.5% of ^{109}Cd , 68.0% of ^{134}Cs , and 71.0% of ^{65}Zn was found on sediments - MP adsorbed <0.8% of these three elements - Adsorption on MP less than sediment reference values	Besson et al. (2020)
2.	^{134}Cs , ^{85}Sr	- HDPE (90–106 μm microspheres) with self-developed biofilms	- Exposure time (142 days) - Estuarine, fresh and marine water - Infrared spectroscopy	- Highest Kd for cesium (Cs) in freshwater (80 mL g^{-1}) and lowest for strontium (Sr) in estuarine conditions (5 mL g^{-1}). - The degree of Cs and Sr adsorption onto plastics appears to be approximately 2–3 orders of magnitude lower than for sediment reference values.	Johansen et al. (2018)
3.	Cd, Co, Cr, Cu, Ni, Pb	- Virgin and aged HDPE plastic pellets	- Exposure time (48 h) - Estuarine conditions	- Beached pellets showed a higher adsorption of all metals compared to virgin pellets. - Differential effects of pH and salinity on the adsorption of different metals. - Non-linear Langmuir or Freundlich isotherms best the data of adsorption	Holmes et al. (2014)
4.	Cu (II), Ni (II), Zn (II), Pd (II)	- Nylon Microplastic particles	- Exposure time (72 h) - Shrinking core model (SCM) used for data fitting - FTIR, BET, XRD, DSC, TGA, and SEM-EDAX - Aqueous with varying pH and Salinity	- Surface O-containing groups on nylon MPs have significant influence on metal adsorption - Langmuir and Freundlich models successfully described the adsorption isotherms - Adsorption kinetics conformed to the pseudo-second order equation, Elovich equation and intraparticle diffusion model well.	Tang et al. (2020)
5.	Fe, Mn, ^{68}Zn , ^{65}Cu , ^{62}Ni , ^{206}Pb , ^{114}Cd	- LDPE, PET, PVC, PS, PP and PA (Nylon) - Both Pristine and aged) - Size (180 μm –3mm, 75–180 μm)	- Seawater (salinity = 30, pH = 8.0) - Lab and Three field sites - Ageing time = 8 Weeks - ICP-MS	- Metal adsorption on aged MP = 2–3 times that of pristine MP - Zn (1.1 $\mu\text{g/g}$) had highest conc. on aged MP - Metal-MP association was related to coprecipitation of Fe and Mn hydroxides followed by metal adsorption on MP	Xie et al. (2021)
6.	Cd, Cr, Cu, Fe, Mn, Ni, Pb, Zn	- Sediment	- Region specific study in Croatia (three beach sites) - Graphite furnace or AAS	- Overall, Fe (88.5 $\mu\text{g/g}$) had highest, while Cd (1.3 ng/g) had lowest concentrations.	Maršić-Lučić et al. (2018)
7.	Cr, Mn, Cu, Zn, As, Cd, Pb	- PVC, PP, PE, PA, PFD - Dia. (4 mm)	- Field (beach) and lab experiments - Room Temp, Exposure time = 16 days - ICP-MS	- More absorption was found on PVC and PP - Max. conc. was found for Mn (31.25 $\mu\text{g/g}$) for field conditions for PP - The type of MP along with the field conditions greatly affects the adsorption of heavy metals.	Gao et al. (2019)
8.	Cd, Cr, Pb, Ag, Cu, Sb, Hg, Fe, Mn	- PE, PP, PS, PA, PVC - Fragments and Fiber MPs	- Case study at three soil sites in China	- The adsorption of heavy metals was directly related to metal contamination at a particular site. - Woodland - 4.1×10^5 particle/kg Vegetable land - 1.6×10^5 particle/kg Vacant Plots - 1.2×10^5 particle/kg	Zhou et al. (2019)
9.	Al, Fe, Mn, Cd, Cr, Ni, Pb, Cu	- MPs (<5 mm) and Sediments	- Field study at nine location along coastline in Persian Gulf - ICP-MS	- Fe had highest concentrations on MPs (531 $\mu\text{g/g}$) and sediments (3050 $\mu\text{g/g}$). - The metal contamination was directly related to the manmade plastic pollution in the adjoining areas.	Doobaradaran et al. (2018)
Organic pollutants					
10.	DDT, Phe, DEHP	- PVC and PE Benthic Invertebrate (fish)	- One-compartment model OMEGA (Optimal Modelling for Ecotoxicological Applications)	- For the benthic invertebrate, predictions including the presence of contaminated plastic resulted in very small increases in the internal concentrations of DDT and DEHP, while the net change in the transfer of Phe was negligible. - Modelling study suggests that ingestion of microplastic does not provide a quantitatively important additional pathway for the transfer of adsorbed chemicals from seawater to biota via the gut.	Bakir et al. (2016)
11.	Dibutyl phthalate (DBP)	- Black plastic film, white plastic film, PE microplastics, 250–600 μm commercial microspheres (CM) 25–1000 μm	- Exposure time: 96 h - Solution used: 0.01 mol L^{-1} CaCl_2 0.2% NaN_3 - SEM, FTIR, DSC, HPLC	- Partition coefficient (Kd) of DBP sorption onto IPPM, BPFM, WPFM and CMs were 1974.55 L kg^{-1} , 1483.85 L kg^{-1} , 1477.45 L/kg and 509.37 L/kg ,	Yao et al. (2021)
12.	Polychlorinated Biphenyl (PCB)	- HDPE MP (10–29 μm) particles with C-14 radiotracer - Sediment particles 1–17.8 μm	- Exposure time: 96 h - Seawater - Liquid scintillation counter, Stereomicroscope	- Plastic particles adsorbed 25.7% of ^{14}C -PCB, while sediment particles adsorbed 89.3% of the same compound. - Microplastic less effective vector for transfer of co-contaminants than the sediment particles	Pyl et al. (2021)

MP = Microplastics, ICP-MS = inductively coupled plasma mass spectrometry, AAS = Atomic Absorption Spectrophotometry.

The association of microplastics with metals has been found to be physical adsorption (electrostatic association), weak coulombic interactions and interparticle diffusion (Tang et al., 2021; Yu et al., 2021). However, non-ionic forms have also been found to be able to interact with the polymers (Holmes et al., 2014). With regards to contaminants, other than metals, the physical adsorption along with absorptive portioning into amorphous domains is the known mechanism (Yao et al., 2021). The kinetics of association (adsorption-desorption) of chemical compounds with the microplastics is still not clear (Besson et al., 2020). However, a study concluded that adsorption of cadmium (II) on polystyrene microplastics (containing hexabromocyclododecane) followed pseudo-second order reaction kinetics (Yu et al., 2021). Further this study revealed that Langmuir isotherm model satisfactorily well described the adsorption. The adsorption kinetics of dibutylphthalate (DBP, a plastic additive) with different microplastics could well be represented by both pseudo-first and second order model, with non-linear Freundlich or Temkin sorption models (Yao et al., 2021). The adsorption kinetics of divalent Cu, Ni and Zn in nylon microplastic followed Elovich and pseudo-second order equation and the Langmuir and Freundlich models successfully described adsorption isotherms (Tang et al., 2021). Moreover, the association of toxic pollutants with the microplastics depends on MP abundance, its scavenging activity and partition coefficient (K_d) in comparison with other media present, residual compounds present in the microplastics (Bakir et al., 2016). In a particular study on cadmium adsorption, the X-ray photon spectroscopy (XPS) revealed that the presence of bromine (due to hexabromocyclododecane in microplastic) significantly adsorbed the metal (Yu et al., 2021). There is lot of discrepancy in comparing the lab-based results and results obtained from field conditions for various trace metals on microplastics and sediments (Besson et al., 2020; Johansen et al., 2018).

Radiotracer approach is cutting edge technology and could be breakthrough for studying the adsorption kinetics of waterborne pollutant-microplastic association with varying microparticle features (size, shape, types) and associated biofilms (Besson et al., 2020). The nuclear and isotope science for studying the virgin microplastic, microplastic associated with metals and other pollutants using radiotracers carbon-13 and hydrogen-2 (deuterium) isotopes, nitrogen-14, and nitrogen-15 isotope after ingestion by marine organisms has been done at International Atomic Energy Agency (IAEA) laboratory, Monaco (Orayeva, 2022). The radiotracers have been used to examine the pathway of microplastics, its detachment from microplastics and leaching into the marine organism. A complete methodological description of sorption of PCB on radiotracer associated microplastic and sediment particles has been described (Pyl et al., 2021). This study gives a deep insight into the use of novel radiotracer technology for the sorption study of co-contaminants on microplastics and sediments.

A comparative study to evaluate the preferential association of three metals including ^{109}Cd , ^{134}Cs , ^{65}Zn with virgin microplastic (PE) and sediment particles in estuarine water has been done. The results revealed after 72 h exposure the adsorption of all the three elements was higher in sediment compared to microplastics. The adsorption of ^{109}Cd , ^{134}Cs , ^{65}Zn after 72 h exposure for sediment and microplastics was $175.17 \pm 15.56 \text{ Bq g}^{-1}$, $721.83 \pm 21.02 \text{ Bq g}^{-1}$, $2474.63 \pm 67.13 \text{ Bq g}^{-1}$ and $17.82 \pm 8.48 \text{ Bq g}^{-1}$, $2.51 \pm 0.47 \text{ Bq g}^{-1}$, $41.16 \pm 4.09 \text{ Bq g}^{-1}$, respectively. The respective K_d of ^{109}Cd , ^{134}Cs , ^{65}Zn for sediment and microplastic were evaluated to be $18.10 \pm 3.14 \text{ mL g}^{-1}$, $96.20 \pm 3.97 \text{ mL g}^{-1}$, $2019.40 \pm 885.28 \text{ mL g}^{-1}$ and $1.70 \pm 1.41 \text{ mL g}^{-1}$, $0.20 \pm 0.04 \text{ mL g}^{-1}$, $3.60 \pm 0.39 \text{ mL g}^{-1}$ (Besson et al., 2020). In another study, same trend on the adsorption of PCB by radiotracer (C-14) associated with HDPE microplastic was observed. The partition coefficient (K_d) for the sediment exceeded 168 times compared to the microplastics (Pyl et al., 2021).

Albeit with heterogeneity, the effect of different plastic and water properties have been studied on the adsorption of metals, co-contaminants and other organic pollutants. (1) Crystallinity: the

crystallinity of microplastic materials play a significant role in the adsorption of co-contaminants and other pollutants (Yao et al., 2021). With increase in crystallinity of the different microplastic materials, the partitioning coefficient of DBP significantly reduced (Yao et al., 2021). (2) Size: the decrease in size amplifies the surface area of microplastic and then increases the adsorption sites for the binding compounds. The microplastic particle size of 300 mesh had maximum adsorption of Cd (II) compared to 35 and 160 mesh sizes at all concentrations ranging from 6 to 16 mg/L (Yu et al., 2021). (3) pH: The increase of pH has found to be increase the adsorption of cadmium (II) on PS (Yu et al., 2021). The range of pH also dictated the mechanism of electrostatic interaction (3–7). The adsorption of Cd, Co, Ni and Pd showed similar trend while Cr and Cu showed reverse trend and no effect, respectively (Holmes et al., 2014). When the pH of a solution is decreased, there is a gradual increase in H^+ ions which compete with the metal to attach to the adsorption sites and H^+ will compete with the metal ions to enter the adsorption sites, thereby reducing the heavy metal adsorption. (4) Salinity: Since the aquatic environment may be different in order of salinity, effect of it on adsorption of metals on microplastics has to be evaluated (Yu et al., 2021). Adsorption as complex function of salinity and highly dependent on metal-plastic-salt ions interaction. The ions of salt and metal compete for adsorption on microplastic besides their electrostatic interaction. With increase in salinity from 0 to 3.5%, the adsorption of cadmium was inhibited. The inhibition was also observed in adsorption of divalent Cu, Ni and Zn on Nylon MP's. Similar trend of adsorption of various metals like cadmium, Nickel, Cu, zinc, cobalt has been observed in other studies (Holmes et al., 2014; Tang et al., 2021). However, the increase in salinity increased the adsorption of Cu, while Cu showed no change of adsorption (Holmes et al., 2014). The partition coefficients (which determine bulk adsorption) of radio cesium and radiostrontium were found more in fresh water than in estuarine conditions (Johansen et al., 2018). With regards to dissolved organic matter (DOM), the adsorption of Cd decreased with an increase in concentration of DOM (Yu et al., 2021). The DOM in the form fulvic acid showed an indifferent effect on the adsorption of divalent Cu, Ni and Zn on Nylon MPs. An increase in FA concentration increased the adsorption of Ni(II) but decreased the adsorption of Cu(II) and Zn(II) adsorption (Tang et al., 2021). The modelling studies of persistent organic pollutants and other hydrophobic organic chemicals have shown negligible association compared to other media present in the aquatic environment (Bakir et al., 2016).

3. Microplastic (MPs) and multiple drug resistance (MDR)

The era of antibiotics has also been called the “Golden Age of Antibiotics” as antibiotics efficiently healed bacterial infections that were previously thought to be treatable (Imran et al., 2019). The discovery of penicillin by Sir Alexander Fleming's in 1928 saved so many lives during World War II and led to antibiotics. Consequently, due to the uncontrolled and inappropriate antibiotic use, the “Golden Age” of antibiotics ended, resulting in the formation of MDR bacteria (Imran et al., 2019). Presently the idea is no longer in effect that antibiotic use or misuse is mainly accountable for the development of AMR, as decreasing and much more regulated antibiotic use did not lead to a significant reduction in the transmission of bacterial resistance (Knapp et al., 2017).

Recently it has been seen that plastic is becoming a matter of concern in both terrestrial and aquatic ecosystems for the development and survival of living organisms (Sathicq et al., 2021a). According to a recent United Nations Environment Program (UNEP) report, seawater will contain more plastic than fish by 2050 (Rocha-Santos, 2018). Plastic has remained in the same state for a long time in the ecosystem. It breaks down into smaller particles less than 5 mm through climate changes and microorganisms' activities (Rummel et al., 2017; Wu et al., 2019). Biofilm is a composite matrix of biodegradable polymers and cellular components formed by microorganisms adhering to surfaces (Dang and Lovell, 2016). In these biofilms, members of species such as *Aeromonas*,

Vibrio, and *E. coli* are generally always prominent (Silva et al., 2019; Viršek et al., 2017).

As a consequence, microplastics are vectors of pathogens in water and other contaminants, and they have now become a hotspot for drug-resistant pathogenic gene transfer (Dong et al., 2021). Meanwhile, MPs particles carrying AMR genes promote genetic contamination, which flows from the sea circulation to the seaside, then from the enriched creatures in the seaside to aquaculture via the food web, where they gradually reach the living organism and inflict unimaginable adverse effects (Dong et al., 2021). In a maricultural environment, the enrichment of antibiotic-resistant bacteria (ARBs), specifically multi-ARBs, on the surface of MPs (polyethylene terephthalate; 75%) was studied (Zhang et al., 2020). Overall, 160 MARB isolates were obtained, with the TET-SFX-ERY-PEN multiple antibiotic resistance profile comprising 25.4% of the microplastic samples and 29% of the samples collected. Antibiotic resistance genes (ARGs) were found in up to 80.0 percent of typical MARB isolates from microplastic samples, and 65.3 percent of MARB isolates from water samples. From this, it clearly indicates that MPs were risky and served as a hotspot for MARB and the emergence of antimicrobial resistance (AMR) (Zhang et al., 2020). Another study reported the possibility of MP harboring and transporting multidrug-resistant *Escherichia coli* throughout various habitats (Song et al., 2020). These type of gram-negative multidrug-resistant microorganisms, particularly *Escherichia coli* generating extended-spectrum beta-lactamase (ESBL), have long become a health problem in aquatic environments (Song et al., 2020). It has been reported that the lateral gene transfer between unicellular and/or multicellular phylogenetically diversified organisms located on microplastics happens substantially faster than in free inhabiting microorganisms (Imran et al., 2019). Meanwhile, other sources, such as waste from industry, mining, agriculture, aquaculture, and sewage, may also have an influence on the environment and have recently been identified as a flashpoint for the emergence of antibiotic-resistant microorganisms (Imran et al., 2019). As a result, MPs are becoming a growing hotspot for antibiotic-polluted marine environments, resistant bacteria, and microplastics, posing a worldwide health concern.

4. Research gaps and future perspectives

Micro (nano)plastics and -resistant genes/multidrug-resistant strains have often been found alone or in combination in both terrestrial and marine culture as emerging pollutants (Dong et al., 2021; Liu et al., 2021). Furthermore, the combined effect of MPs and ARGs causes gene dispersion and cross-pollution in the environment, endangering aquaculture and human health (Dong et al., 2021). However, the results of the reported studies so far are in the speculative phase, which shows that the micro (nano)plastics (MNPs) has not yet been thoroughly researched in context with MDR, due to which it is not right to draw any conclusions about its vector potential and spread of antibiotic-resistant. As a result, future research and studies on the impacts of microorganisms habitats on MNPs, gene exchange among species (ARGs), adsorption of MPs, diffusion mechanism of pathogenic organisms over MNPs, identification of important microbial communities, species diversity in various habitats (terrestrial and aquatic), and an environmental health risk index system of MNPs pollution are all recommended (Shen et al., 2019).

Depending on the chemical and physical features of micro (nano) plastics, they can have varied impacts on microbiota and activity, and ultimately on the systemic ecosystem (Amalfitano et al., 2017; Enke et al., 2018). Heterogenous microplastic pollution provides a plethora of new ecological and geographical footprints for a variety of potentially harmful AMR microorganisms that cannot survive in their natural habitat. Presently the role of MPs in adapting microbial diversity is based on studies of a few conventional plastic particles such as polypropylene, polyethylene, and polyethylene terephthalate, whereas globally, a range of several polymers and organic compositions represent an accountable proportion of MNPs pollution; therefore, in this

context, there is a need for comprehensive research. For instance, road traffic-generated tyres and brake wear particles are among the most commonly observed MPs worldwide, although its plastisphere remains mostly unknown (Evangelidou et al., 2020). In addition, the biodegradable plastics, which have increased as an appropriate alternative to the conventional hydrocarbon-based polymer-plastics due to their biodegradability, renewable origin, and application areas, have mainly been well studied as an environmental pollutant that might also interfere with natural microorganisms (González-Pleiter et al., 2019; Kumar et al., 2018). Until, only a few experimental investigations have looked into the potential effects of biodegradable MNPs on water species (González-Pleiter et al., 2019), suggesting the need for research on marine and freshwater species. The diversification of microbial population cultured on polyhydroxyalkanoate (PHA) or polyethylene (PE) has been studied, and it was shown that sulfate-reducing bacteria were substantially prevalent on PHA (Pinnell and Turner, 2019). However, their environmental and bacteriological relevance is unknown (Shruti and Kutralam-Muniasamy, 2019). Therefore, comprehensive knowledge of MPs is required for efficient monitoring of their influence on the environment, considering their composition, origin, form, and environmental, evolutionary, and toxicological role within a known natural ecosystem mainly when it takes into consideration of organisms and genes (Sathicq et al., 2021b). Furthermore, a better knowledge of the function of environmental conditions in the enrichment of ARGs on MPs (Liu et al., 2021) and its role in the transit of ARGs throughout various food webs is still unknown and requires further studies. To summarize, thorough research is necessary to determine the precise mechanism underlying the relationship between microbial population and MPs and their synergistic or antagonistic behavior to the environment, which is critical for understanding the possible harm of ARGs on microplastics.

5. Conclusions

Microplastics and antibiotics has become a global concern owing to their persistence and adverse effect on human and environmental health. Although as distant yet these priority pollutants have significant correlation. The occurrence of microplastics has served as a host for the proliferation of adverse effects caused by antibiotics i.e. antimicrobial or multidrug resistance. In order to correctly address the problem, it's imperative to study it objectively by identifying the factors involved. Therefore, the co-study of microplastics with antimicrobial resistance emergence can be helpful for researchers globally to find efficient tool to combat this problem.

Credit author statement

Vivek Kumar Gaur: Literature survey, Writing- Original draft preparation; **Ranjna Sirohi:** Literature survey, Writing- Original draft preparation; **Mohd Ishfaq Bhat:** Literature survey, Writing- Original draft preparation; **Krishna Gautam:** Literature survey, Writing- Original draft preparation; **Poonam Sharma:** Literature survey, Writing- Original draft preparation; **Janmejai Kumar Srivastava:** Supervision; Writing-Reviewing and Editing **Ashok Pandey:** Conceptualization, Supervision; Writing-Reviewing and Editing;

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

This manuscript bears the Integral University communication number IU/R&D/2022- MCN0001617.

References

- Alabi, O.A., Ologbonjaye, K.I., Awosolu, O., Alalade, O.E., 2019. Public and environmental health effects of plastic wastes disposal: a review. *J. Toxicol. Risk Assess.* 5, 1–13.
- Allen, S., Allen, D., Moss, K., le Roux, G., Phoenix, V.R., Sonke, J.E., 2020. Examination of the ocean as a source for atmospheric microplastics. *PLoS One* 15, e0232746.
- Allouzi, M.M.A., Tang, D.Y.Y., Chew, K.W., Rinklebe, J., Bolan, N., Allouzi, S.M.A., Show, P.L., 2021. Micro (nano) plastic pollution: the ecological influence on soil-plant system and human health. *Sci. Total Environ.* <https://doi.org/10.1016/j.scitotenv.2021.147815>.
- Alshehrei, F., 2017. Biodegradation of synthetic and natural plastic by microorganisms. *J. Appl. Environ. Microbiol.* 5, 8–19.
- Amalfitano, S., Corno, G., Eckert, E., Fazi, S., Ninio, S., Callieri, C., Grossart, H.P., Eckert, W., 2017. Tracing particulate matter and associated microorganisms in freshwaters. *Hydrobiologia.* <https://doi.org/10.1007/s10750-017-3260-x>.
- Amato-Lourenço, L.F., Costa, N.D.S.X., Dantas, K.C., dos Santos Galvão, L., Morales, F. N., Lombardi, S.C.F.S., Júnior, A.M., Lindoso, J.A.L., Ando, R.A., Lima, F.G., Carvalho-Oliveira, R., 2022. Airborne microplastics and SARS-CoV-2 in total suspended particles in the area surrounding the largest medical centre in Latin America. *Environmental Pollution* 292, 118299.
- Andrews, A., Budd, E.L., Hendrick, A., Ashiru-Oredope, D., Beech, E., Hopkins, S., Gerver, S., Muller-Pebody, B., 2021. Surveillance of antibacterial usage during the COVID-19 pandemic in England, 2020. *Antibiotics.* <https://doi.org/10.3390/antibiotics10070841>.
- Arias-Andres, M., Klümper, U., Rojas-Jimenez, K., Grossart, H.P., 2018. Microplastic pollution increases gene exchange in aquatic ecosystems. *Environ. Pollut.* <https://doi.org/10.1016/j.envpol.2018.02.058>.
- Ashton, K., Holmes, L., Turner, A., 2010. Association of metals with plastic production pellets in the marine environment. *Mar. Pollut. Bull.* 60, 2050–2055.
- Azeem, I., Adeel, M., Ahmad, M.A., Shakoor, N., Jiangcuo, G.D., Azeem, K., Ishfaq, M., Shakoor, A., Ayaz, M., Xu, M., 2021. Uptake and accumulation of nano/microplastics in plants: a critical review. *Nanomaterials* 11, 2935.
- Bakir, A., O'Connor, I.A., Rowland, S.J., Hendriks, A.J., Thompson, R.C., 2016. Relative importance of microplastics as a pathway for the transfer of hydrophobic organic chemicals to marine life. *Environ. Pollut.* 219, 56–65.
- Bartkova, S., Kahru, A., Heinlaan, M., Scheler, O., 2021. Techniques used for analyzing microplastics, antimicrobial resistance and microbial community composition: a mini-review. *Front. Microbiol.* 12, 413.
- Besseling, E., Redondo-Hasselerharm, P., Foekema, E.M., Koelmans, A.A., 2019. Quantifying ecological risks of aquatic micro- and nanoplastic. *Crit. Rev. Environ. Sci. Technol.* <https://doi.org/10.1080/10643389.2018.1531688>.
- Besson, M., Jacob, H., Oberhaensli, F., Taylor, A., Swarzenski, P.W., Metian, M., 2020. Preferential adsorption of Cd, Cs and Zn onto virgin polyethylene microplastic versus sediment particles. *Mar. Pollut. Bull.* 156, 111223.
- Blakely, G.W., 2014. Mechanisms of horizontal gene transfer and DNA recombination. In: *Molecular Medical Microbiology*, second ed. <https://doi.org/10.1016/B978-0-12-397169-2.00015-9>.
- Boyle, K., Örmeci, B., 2020. Microplastics and nanoplastics in the freshwater and terrestrial environment: a review. *Water* 2020 12. <https://doi.org/10.3390/W12092633>. Page 2633 12, 2633.
- Campanale, C., Massarelli, C., Savino, I., Locaputo, V., Uricchio, V.F., 2020. A detailed review study on potential effects of microplastics and additives of concern on human health. *Int. J. Environ. Res. Publ. Health* 17, 1212.
- Center for Disease Control and Prevention, 2019. Antibiotic Resistance Threats in the United States, 2019. U.S. Department of Health and Human Services. Center for Disease control and Prevention, Atlanta, GA.
- Chaturvedi, P., Giri, B.S., Shukla, P., Gupta, P., 2021. Recent advancement in remediation of synthetic organic antibiotics from environmental matrices: challenges and perspective. *Bioresour. Technol.* 319, 124161 <https://doi.org/10.1016/j.biortech.2020.124161>.
- Corcoran, P.L., Moore, C.J., Jazvac, K., 2014. An anthropogenic marker horizon in the future rock record. *GSA Today (Geol. Soc. Am.)* 24, 4–8. <https://doi.org/10.1130/GSAT-G198A.1>.
- Courtene-Jones, W., Quinn, B., Gary, S.F., Mogg, A.O.M., Narayanaswamy, B.E., 2017. Microplastic pollution identified in deep-sea water and ingested by benthic invertebrates in the Rockall Trough, North Atlantic Ocean. *Environ. Pollut.* 231, 271–280.
- Cui, G., Lü, F., Zhang, H., Shao, L., He, P., 2020. Critical insight into the fate of antibiotic resistance genes during biological treatment of typical biowastes. *Bioresour. Technol.* 317, 123974 <https://doi.org/10.1016/j.biortech.2020.123974>.
- Cui, Y., Gao, J., Zhang, D., Li, D., Dai, H., Wang, Z., Zhao, Y., 2021. Responses of performance, antibiotic resistance genes and bacterial communities of partial nitrification system to polyamide microplastics. *Bioresour. Technol.* 341, 125767 <https://doi.org/10.1016/j.biortech.2021.125767>.
- Dadgostar, P., 2019. Antimicrobial resistance: implications and costs. *Infect. Drug Resist.* <https://doi.org/10.2147/IDR.S234610>.
- Dang, H., Lovell, C.R., 2016. Microbial surface colonization and biofilm development in marine environments. *Microbiol. Mol. Biol. Rev.* <https://doi.org/10.1128/mmr.00037-15>.
- de Kraker, M.E.A., Stewardson, A.J., Harbarth, S., 2016. Will 10 million people die a year due to antimicrobial resistance by 2050? *PLoS Med.* <https://doi.org/10.1371/journal.pmed.1002184>.
- De-la-Torre, G.E., Dioses-Salinas, D.C., Pizarro-Ortega, C.I., Santillán, L., 2021. New plastic formations in the Anthropocene. *Sci. Total Environ.* 754, 142216.
- Dobaradaran, S., Schmidt, T.C., Nabipour, I., Khajeahmadi, N., Tajbakhsh, S., Saeedi, R., Javad Mohammadi, M., Keshkar, M., Khorsand, M., Faraji Ghasemi, F., 2018. Characterization of plastic debris and association of metals with microplastics in coastline sediment along the Persian Gulf. *Waste Manag.* 78, 649–658. <https://doi.org/10.1016/j.wasman.2018.06.037>.
- Dong, H., Chen, Y., Wang, J., Zhang, Y., Zhang, P., Li, X., Zou, J., Zhou, A., 2021. Interactions of microplastics and antibiotic resistance genes and their effects on the aquaculture environments. *J. Hazard Mater.* <https://doi.org/10.1016/j.jhazmat.2020.123961>.
- Duval, R.E., Grare, M., Demoré, B., 2019. Fight against antimicrobial resistance: we always need new antibacterials but for right bacteria. *Molecules* 24, 3152.
- ECDC, 2020. European Centre for Disease Prevention and Control. Antimicrobial Resistance in the EU/EEA (EARS-Net) - Annual Epidemiological Report 2019. Centre for Disease Control and Prevention.
- Ehlers, S.M., Ellrich, J.A., 2020. First record of “plasticrusts” and “pyroplastic” from the Mediterranean Sea. *Mar. Pollut. Bull.* 151 <https://doi.org/10.1016/j.marpolbul.2019.110845>.
- Elabbadi, A., Turpin, M., Gerotziakas, G.T., Teulier, M., Voiriot, G., Fartoukh, M., 2021. Bacterial co-infection in critically ill COVID-19 patients with severe pneumonia. *Infection.* <https://doi.org/10.1007/s15010-020-01553-x>.
- Enke, T.N., Leventhal, G.E., Metzger, M., Saavedra, J.T., Cordero, O.X., 2018. Microscale ecology regulates particulate organic matter turnover in model marine microbial communities. *Nat. Commun.* <https://doi.org/10.1038/s41467-018-05159-8>.
- Essack, S., 2018. Antibiotic resistance and One Health: a mapping project. *Lancet Global Health.* [https://doi.org/10.1016/s2214-109x\(18\)30156-6](https://doi.org/10.1016/s2214-109x(18)30156-6).
- Evangelidou, N., Grythe, H., Klimont, Z., Heyes, C., Eckhardt, S., Lopez-Aparicio, S., Stohl, A., 2020. Atmospheric transport is a major pathway of microplastics to remote regions. *Nat. Commun.* <https://doi.org/10.1038/s41467-020-17201-9>.
- Fernandino, G., Elliff, C.I., Franciscini, H., Dentzien-Dias, P., 2020. Anthroplastics: first description of plastics and other man-made materials in recently formed coastal sedimentary rocks in the southern hemisphere. *Mar. Pollut. Bull.* 154, 111044.
- Fred-Ahmadu, O.H., Bhagwat, G., Oluyoye, I., Benson, N.U., Ayejuyi, O.O., Palanisami, T., 2020. Interaction of chemical contaminants with microplastics: principles and perspectives. *Sci. Total Environ.* 706, 135978.
- Frère, L., Maignien, L., Chalopin, M., Huvet, A., Rinnert, E., Morrison, H., Kerninon, S., Cassone, A.L., Lambert, C., Reveillaud, J., Paul-Pont, I., 2018. Microplastic bacterial communities in the Bay of Brest: influence of polymer type and size. *Environ. Pollut.* <https://doi.org/10.1016/j.envpol.2018.07.023>.
- Frias, J.P.G.L., Nash, R., 2019. Microplastics: finding a consensus on the definition. *Mar. Pollut. Bull.* <https://doi.org/10.1016/j.marpolbul.2018.11.022>.
- Gao, F., Li, J., Sun, C., Zhang, L., Jiang, F., Cao, W., Zheng, L., 2019. Study on the capability and characteristics of heavy metals enriched on microplastics in marine environment. *Mar. Pollut. Bull.* 144, 61–67. <https://doi.org/10.1016/j.marpolbul.2019.04.039>.
- Gao, D., Liu, X., Junaid, M., Liao, H., Chen, G., Wu, Y., Wang, J., 2022. Toxicological impacts of micro(nano)plastics in the benthic environment. *Sci. Total Environ.* 836 <https://doi.org/10.1016/j.scitotenv.2022.155620>.
- Gaur, V.K., Gupta, S., Sharma, P., Gupta, P., Varjani, S., Srivastava, J.K., Chang, J.S., Bui, X.T., 2022. Metabolic cascade for remediation of plastic waste: a case study on microplastic degradation. *Curr. Pollut. Rep.* 8, 30–50. <https://doi.org/10.1007/S40726-021-00210-7/TABLES/4>.
- Gerritse, J., Leslie, H.A., de Tender, C.A., Devriese, L.I., Vethaak, A.D., 2020. Fragmentation of plastic objects in a laboratory seawater microcosm. *Sci. Rep.* 10, 1–16.
- Gestoso, L., Cacabelos, E., Ramalhosa, P., Canning-Clode, J., 2019. Plasticrusts: a new potential threat in the Anthropocene's rocky shores. *Sci. Total Environ.* 687, 413–415.
- Gigault, J., Halle, A. ter, Baudrimont, M., Pascal, P.Y., Gauffre, F., Phi, T.L., El Hadri, H., Grassl, B., Reynaud, S., 2018. Current opinion: what is a nanoplastic? *Environ. Pollut.* <https://doi.org/10.1016/j.envpol.2018.01.024>.
- Gkoutselis, G., Rohrbach, S., Harjes, J., Obst, M., Brachmann, A., Horn, M.A., Rambold, G., 2021. Microplastics accumulate fungal pathogens in terrestrial ecosystems. *Sci. Rep.* <https://doi.org/10.1038/s41598-021-92405-7>, 2021 11:1 11, 1–13.
- González-Pleiter, M., Tamayo-Belda, M., Pulido-Reyes, G., Amariei, G., Leganés, F., Rosal, R., Fernández-Piñas, F., 2019. Secondary nanoplastics released from a biodegradable microplastic severely impact freshwater environments. *Environ. Sci. Nano.* <https://doi.org/10.1039/c8en01427b>.
- Haegerbaeumer, A., Mueller, M.T., Fueser, H., Traunsperger, W., 2019. Impacts of micro- and nano-sized plastic particles on benthic invertebrates: a literature review and gap analysis. *Front. Environ. Sci.* <https://doi.org/10.3389/fenvs.2019.00017>.
- He, Z.W., Yang, W.J., Ren, Y.X., Jin, H.Y., Tang, C.C., Liu, W.Z., Yang, C.X., Zhou, A.J., Wang, A.J., 2021. Occurrence, effect, and fate of residual microplastics in anaerobic digestion of waste activated sludge: a state-of-the-art review. *Bioresour. Technol.* 331, 125035 <https://doi.org/10.1016/j.biortech.2021.125035>.
- Holmes, L.A., Turner, A., Thompson, R.C., 2012. Adsorption of trace metals to plastic resin pellets in the marine environment. *Environ. Pollut.* 160, 42–48.

- Holmes, L.A., Turner, A., Thompson, R.C., 2014. Interactions between trace metals and plastic production pellets under estuarine conditions. *Mar. Chem.* 167, 25–32.
- Huang, F.-Y., Yang, K., Zhang, Z.-X., Su, J.-Q., Zhu, Y.-G., Zhang, X., 2019. Effects of microplastics on antibiotic resistance genes in estuarine sediments. *Huanjing Kexue* 40, 2234–2239.
- Hui, D.S., I Azhar, E., Madani, T.A., Ntoumi, F., Kock, R., Dar, O., Ippolito, G., Mchugh, T. D., Memish, Z.A., Drosten, C., Zumla, A., Petersen, E., 2020. The continuing 2019-nCoV epidemic threat of novel coronaviruses to global health — the latest 2019 novel coronavirus outbreak in Wuhan, China. *Int. J. Infect. Dis.* <https://doi.org/10.1016/j.ijid.2020.01.009>.
- Huttner, B.D., Catho, G., Pano-Pardo, J.R., Pulcini, C., Schouten, J., 2020. COVID-19: Don't Neglect Antimicrobial Stewardship Principles! *Clinical Microbiology and Infection.* <https://doi.org/10.1016/j.cmi.2020.04.024>.
- Imran, M., Das, K.R., Naik, M.M., 2019. Co-selection of multi-antibiotic resistance in bacterial pathogens in metal and microplastic contaminated environments: an emerging health threat. *Chemosphere.* <https://doi.org/10.1016/j.chemosphere.2018.10.114>.
- Johansen, M.P., Prentice, E., Cresswell, T., Howell, N., 2018. Initial data on adsorption of Cs and Sr to the surfaces of microplastics with biofilm. *J. Environ. Radioact.* 190, 130–133.
- Joo, S.H., Liang, Y., Kim, M., Byun, J., Choi, H., 2021. Microplastics with adsorbed contaminants: mechanisms and treatment. *Environ. Chall.* 3, 100042.
- Karkman, A., Do, T.T., Walsh, F., Virta, M.P.J., 2018. Antibiotic-resistance genes in waste water. *Trends Microbiol.* <https://doi.org/10.1016/j.tim.2017.09.005>.
- Kaza, S., Yao, L., Bhada-Tata, P., van Woerden, F., 2018. What a Waste 2.0: a Global Snapshot of Solid Waste Management to 2050. World Bank Publications.
- Knapp, C.W., Callan, A.C., Aitken, B., Shearn, R., Koenders, A., Hinwood, A., 2017. Relationship between antibiotic resistance genes and metals in residential soil samples from Western Australia. *Environ. Sci. Pollut. Control Ser.* <https://doi.org/10.1007/s11356-016-7997-y>.
- Knight, G.M., Glover, R.E., McQuaid, C.F., Olaru, I.D., Gallandat, K., Leclerc, Q.J., Fuller, N.M., Willcocks, S.J., Hasan, R., van Kleef, E., Chandler, C.I.R., 2021. Antimicrobial Resistance and Covid-19: Intersections and Implications. *Elife.* <https://doi.org/10.7554/eLife.64139>.
- Kumar, V., Thakur, V., Ambika Kumar, S., Singh, D., 2018. Bioplastic reservoir of diverse bacterial communities revealed along altitude gradient of Pangi-Chamba trans-Himalayan region. *FEMS Microbiol. Lett.* <https://doi.org/10.1093/femsle/fny144>.
- Kumar, M., Chen, H., Sarsaiya, S., Qin, S., Liu, H., Awasthi, M.K., Kumar, S., Singh, L., Zhang, Z., Bolan, N.S., Pandey, A., Varjani, S., Taherzadeh, M.J., 2021. Current research trends on micro- and nano-plastics as an emerging threat to global environment: a review. *J. Hazard Mater.* <https://doi.org/10.1016/j.jhazmat.2020.124967>.
- Langford, B.J., So, M., Raybardhan, S., Leung, V., Westwood, D., MacFadden, D.R., Soucy, J.P.R., Daneman, N., 2020. Bacterial co-infection and secondary infection in patients with COVID-19: a living rapid review and meta-analysis. *Clin. Microbiol. Infect.* <https://doi.org/10.1016/j.cmi.2020.07.016>.
- Langford, B.J., So, M., Raybardhan, S., Leung, V., Soucy, J.P.R., Westwood, D., Daneman, N., MacFadden, D.R., 2021. Antibiotic prescribing in patients with COVID-19: rapid review and meta-analysis. *Clin. Microbiol. Infect.* <https://doi.org/10.1016/j.cmi.2020.12.018>.
- Larsson, D.G.J., Flach, C.F., 2021. Antibiotic resistance in the environment. *Nat. Rev. Microbiol.* <https://doi.org/10.1038/s41579-021-00649-x>.
- Leal Filho, W., Saari, U., Fedoruk, M., Iital, A., Moora, H., Klöga, M., Voronova, V., 2019. An overview of the problems posed by plastic products and the role of extended producer responsibility in Europe. *J. Clean. Prod.* 214, 550–558.
- Li, J., Zhang, K., Zhang, H., 2018. Adsorption of antibiotics on microplastics. *Environ. Pollut.* 237, 460–467. <https://doi.org/10.1016/j.envpol.2018.02.050>.
- Liu, Q., Schauer, J., 2021. Airborne microplastics from waste as a transmission vector for COVID-19. *Aerosol Air Qual. Res.* 21, 200439 <https://doi.org/10.4209/AAQR.2020.07.0439>.
- Liu, Y., Liu, W., Yang, X., Wang, J., Lin, H., Yuyi, Y., 2021. Microplastics are a hotspot for antibiotic resistance genes: progress and perspective. *Sci. Total Environ.* 145643.
- Liu, P., Dai, J., Huang, K., Yang, Z., Zhang, Z., Guo, X., 2022. Sources of micro(nano) plastics and interaction with co-existing pollutants in wastewater treatment plants. *Crit. Rev. Environ. Sci. Technol.* 1–21 https://doi.org/10.1080/10643389.2022.2095844/SUPPL_FILE/BEST_A_2095844_SM7043.DOCX.
- Lobie, T.A., Roba, A.A., Booth, J.A., Kristiansen, K.I., Aseffa, A., Skarstad, K., Bjørås, M., 2021. Antimicrobial resistance: a challenge awaiting the post-COVID-19 era. *Int. J. Infect. Dis.* <https://doi.org/10.1016/j.ijid.2021.09.003>.
- López-Velandia, D.P., Torres-Caycedo, M.I., Castro-Gutiérrez, L.T., Prada-Quiroga, C.F., 2018. Antibiotic resistance: origins, evolution and healthcare-associated infections. *Salud Uninorte.* <https://doi.org/10.14482/sun.34.2.615.32>.
- Lu, J., Zhang, Y., Wu, J., Luo, Y., 2019. Effects of microplastics on distribution of antibiotic resistance genes in recirculating aquaculture system. *Ecotoxicol. Environ. Saf.* 184, 109631.
- Lusher, A., 2015. Microplastics in the marine environment: distribution, interactions and effects. In: *Marine Anthropogenic Litter.* https://doi.org/10.1007/978-3-319-16510-3_10.
- Ma, J., Sheng, G.D., O'Connor, P., 2020. Microplastics combined with tetracycline in soils facilitate the formation of antibiotic resistance in the *Enchytraeus crypticus* microbiome. *Environ. Pollut.* 264, 114689 <https://doi.org/10.1016/j.envpol.2020.114689>.
- Ma, J., Gong, Z., Wang, Z., Liu, H., Chen, G., Guo, G., 2022. Elucidating degradation properties, microbial community, and mechanism of microplastics in sewage sludge under different terminal electron acceptors conditions. *Bioresour. Technol.* 346, 126624 <https://doi.org/10.1016/j.biortech.2021.126624>.
- Malizia, A., Monmany-Garzia, A.C., 2019. Terrestrial ecologists should stop ignoring plastic pollution in the Anthropocene time. *Sci. Total Environ.* 668, 1025–1029.
- Mantlo, E., Rhodes, T., Boutros, J., Patterson-Fortin, L., Evans, A., Paessler, S., 2020. In Vitro Efficacy of a Copper Iodine Complex PPE Disinfectant for SARS-CoV-2 Inactivation. <https://doi.org/10.12688/f1000research.24651.1>. F1000Res.
- Marsić-Lučić, J., Lušić, J., Tutman, P., Bojanić Varezić, D., Šiljić, J., Pribudić, J., 2018. Levels of trace metals on microplastic particles in beach sediments of the island of Vis, Adriatic Sea, Croatia. *Mar. Pollut. Bull.* 137, 231–236. <https://doi.org/10.1016/j.marpolbul.2018.10.027>.
- Mohammad Mirsoleimani Azizi, S., Hai, F.I., Lu, W., Al-Mamun, A., Ranjan Dhar, B., 2021. A review of mechanisms underlying the impacts of (nano)microplastics on anaerobic digestion. *Bioresour. Technol.* 329, 124894 <https://doi.org/10.1016/j.biortech.2021.124894>.
- Murray, A.K., 2020. The novel coronavirus COVID-19 outbreak: global implications for antimicrobial resistance. *Front. Microbiol.* <https://doi.org/10.3389/fmicb.2020.01020>.
- Mytum, H., Meek, J., 2021. The Iron age in the plastic age: Anthropocene signatures at castell henllys. *Antiquity* 95, 198–214.
- Nayak, B., Parshionikar, S., 2021. Beware of Superbugs in a Post-COVID World. *J Am Water Works Assoc.* <https://doi.org/10.1002/awwa.1713>.
- Nguyen, B., Claveau-Mallet, D., Hernandez, L.M., Xu, E.G., Farner, J.M., Tufenkji, N., 2019. Separation and analysis of microplastics and nanoplastics in complex environmental samples. *Acc. Chem. Res.* 52, 858–866. <https://doi.org/10.1021/acs.accounts.8b00602>.
- Oberbeckmann, S., Löder, M.G.J., Labrenz, M., 2015. Marine microplastic-associated biofilms - a review. *Environ. Chem.* <https://doi.org/10.1071/EN15069>.
- Orayeva, J., 2022. Tracking the Impact of Plastics Pollution on Marine Life.
- Pinnell, L.J., Turner, J.W., 2019. Shotgun metagenomics reveals the benthic microbial community response to plastic and bioplastic in a coastal marine environment. *Front. Microbiol.* <https://doi.org/10.3389/fmicb.2019.01252>.
- Pires, D., De Kraker, M.E.A., Tartari, E., Abbas, M., Pittet, D., 2017. In: *Fight Antibiotic Resistance - It's in Your Hands: Call from the World Health Organization for 5th May 2017* <https://doi.org/10.1093/cid/cix226>. *Clinical Infectious Diseases.*
- Porta, R., 2021. Anthropocene, the plastic age and future perspectives. *FEBS Open Bio* 4, 948–953.
- Prestinaci, P., Pezzotti, P., Pantosti, A., 2015. Antimicrobial resistance: a global multifaceted phenomenon. *Pathog. Glob. Health.* <https://doi.org/10.1179/2047773215Y.0000000030>.
- Prüst, M., Meijer, J., Westerink, R.H.S., 2020. The plastic brain: neurotoxicity of micro- and nanoplastics. *Part. Fibre Toxicol.* <https://doi.org/10.1186/s12989-020-00358-y>.
- Pyl, M., Danis, B., Oberhaensli, F., Taylor, A., Swarzenski, P., Metian, M., 2021. An effective method to assess the sorption dynamics of PCB radiotracers onto plastic and sediment microparticles. *MethodsX* 8, 101395.
- Rai, P.K., Lee, J., Brown, R.J.C., Kim, K.-H., 2021. Micro- and nano-plastic pollution: behavior, microbial ecology, and remediation technologies. *J. Clean. Prod.* 291, 125240.
- Rawson, T.M., Moore, L.S.P., Zhu, N., Ranganathan, N., Skolimowska, K., Gilchrist, M., Satta, G., Cooke, G., Holmes, A., 2020. Bacterial and fungal coinfection in individuals with coronavirus: a rapid review to support COVID-19 antimicrobial prescribing. *Clin. Infect. Dis.* <https://doi.org/10.1093/cid/ciaa530>.
- Rinanda, T., Fathima, R., Ramadhani, R., 2019. Microbiological perspectives on the effects of microplastics on the aquatic environment. In: *IOP Conference Series: Earth and Environmental Science.* IOP Publishing, 012048.
- Rizvi, S.G., Ahammad, S.Z., 2022. COVID-19 and antimicrobial resistance: a cross-study. *Sci. Total Environ.* 807, 150873 <https://doi.org/10.1016/j.scitotenv.2021.150873>.
- Rocha-Santos, T.A.P., 2018. Editorial overview: micro and nano-plastics. *Curr. Opin Environ. Sci. Health.* <https://doi.org/10.1016/j.coesh.2018.01.003>.
- Rodríguez-Baño, J., Rossolini, G.M., Schultz, C., Tacconelli, E., Murthy, S., Ohmagari, N., Holmes, A., Bachmann, T., Goossens, H., Canton, R., Roberts, A.P., Henriques-Normark, B., Clancy, C.J., Huttner, B., Fagerstedt, P., Lahiri, S., Kaushic, C., Hoffman, S.J., Warren, M., Zoubiane, G., Essack, S., Laxminarayan, R., Plant, L., 2021. Antimicrobial resistance research in a post-pandemic world: insights on antimicrobial resistance research in the COVID-19 pandemic. *J. Glob. Antimicrob. Resist.* <https://doi.org/10.1016/j.jgar.2021.02.013>.
- Romeo, J., 2020. Do No Harm: plastics are playing a major role in giving healthcare professionals the tools and capabilities they need to battle the COVID pandemic. *Plast. Eng.* 76, 41.
- Ruan, Y., Zhang, X., Qiu, J.-W., Leung, K.M.Y., Lam, J.C.W., Lam, P.K.S., 2018. Stereoisomer-specific trophodynamics of the chiral brominated flame retardants HBCD and TBEC in a marine food web, with implications for human exposure. *Environ. Sci. Technol.* 52, 8183–8193.
- Rummel, C.D., Jahnke, A., Gorokhova, E., Kühnel, D., Schmitt-Jansen, M., 2017. Impacts of biofilm formation on the fate and potential effects of microplastic in the aquatic environment. *Environ. Sci. Technol. Lett.* <https://doi.org/10.1021/acs.estlett.7b00164>.
- Saini, V., Jain, C., Singh, N.P., Alsulimani, A., Gupta, C., Dar, S.A., Haque, S., Das, S., 2021. Paradigm shift in antimicrobial resistance pattern of bacterial isolates during the covid-19 pandemic. *Antibiotics.* <https://doi.org/10.3390/antibiotics10080954>.
- Samreen, Ahmad, I., Malak, H.A., Abulreesh, H.H., 2021. Environmental antimicrobial resistance and its drivers: a potential threat to public health. *J. Glob. Antimicrob. Resist.* <https://doi.org/10.1016/j.jgar.2021.08.001>.
- Sathicq, M.B., Sabatino, R., Corno, G., di Cesare, A., 2021a. Are microplastic particles a hotspot for the spread and the persistence of antibiotic resistance in aquatic systems? *Environ. Pollut.* 279, 116896 <https://doi.org/10.1016/j.envpol.2021.116896>.

- Sathicq, M.B., Sabatino, R., Corno, G., Di Cesare, A., 2021b. Are microplastic particles a hotspot for the spread and the persistence of antibiotic resistance in aquatic systems? *Environ. Pollut.* <https://doi.org/10.1016/j.envpol.2021.116896>.
- Sawatwong, P., Sapchookul, P., Whistler, T., Gregory, C.J., Sangwichian, O., Makprasert, S., Jorakate, P., Srisaengchai, P., Thamthitiwat, S., Promkong, C., Nanvattanachod, P., Vanaporn, M., Rhodes, J., 2019. High burden of extended-spectrum β -lactamase-producing *Escherichia coli* and *klebsiella pneumoniae* bacteremia in older adults: a seven-year study in two rural Thai Provinces. *Am. J. Trop. Med. Hyg.* <https://doi.org/10.4269/ajtmh.18-0394>.
- Shen, M., Zhu, Y., Zhang, Y., Zeng, G., Wen, X., Yi, H., Ye, S., Ren, X., Song, B., 2019. Micro(nano)plastics: unignorable vectors for organisms. *Mar. Pollut. Bull.* <https://doi.org/10.1016/j.marpolbul.2019.01.004>.
- Shruti, V.C., Kuttralam-Muniasamy, G., 2019. Bioplastics: missing link in the era of microplastics. *Sci. Total Environ.* <https://doi.org/10.1016/j.scitotenv.2019.134139>.
- e Silva, P.P.G., Nobre, C.R., Resaffe, P., Pereira, C.D.S., Gusmão, F., 2016. Leachate from microplastics impairs larval development in brown mussels. *Water Res.* 106, 364–370.
- Silva, M.M., Maldonado, G.C., Castro, R.O., de Sá Felizardo, J., Cardoso, R.P., Anjos, R. M., dos, Araújo, F.V. de, 2019. Dispersal of potentially pathogenic bacteria by plastic debris in Guanabara Bay, RJ, Brazil. *Mar. Pollut. Bull.* <https://doi.org/10.1016/j.marpolbul.2019.02.064>.
- Song, J., Jongmans-Hochschulz, E., Mauder, N., Imirzalioglu, C., Wichels, A., Gerdt, G., 2020. The Travelling Particles: Investigating Microplastics as Possible Transport Vectors for Multidrug Resistant *E. coli* in the Weser Estuary (Germany). *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2020.137603>.
- Su, Y., Zhang, Z., Zhu, J., Shi, J., Wei, H., Xie, B., Shi, H., 2021. Microplastics act as vectors for antibiotic resistance genes in landfill leachate: the enhanced roles of the long-term aging process. *Environ. Pollut.* 270, 116278 <https://doi.org/10.1016/j.envpol.2020.116278>.
- Sun, Yuanze, Cao, N., Duan, C., Wang, Q., Ding, C., Wang, J., 2021a. Selection of antibiotic resistance genes on biodegradable and non-biodegradable microplastics. *J. Hazard Mater.* <https://doi.org/10.1016/j.jhazmat.2020.124979>.
- Sun, Yue, Ren, X., Rene, E.R., Wang, Z., Zhou, L., Zhang, Z., Wang, Q., 2021b. The degradation performance of different microplastics and their effect on microbial community during composting process. *Bioresour. Technol.* 332, 125133 <https://doi.org/10.1016/j.biortech.2021.125133>.
- Taffel, S., 2016. Technofossils of the Anthropocene: media, geology, and plastics. *Cult. Polit.* 12, 355–375.
- Tamminga, M., Hengstmann, E., Fischer, E.K., 2018. Microplastic analysis in the South Funen Archipelago, Baltic Sea, implementing manta trawling and bulk sampling. *Mar. Pollut. Bull.* 128, 601–608. <https://doi.org/10.1016/j.marpolbul.2018.01.066>.
- Tang, S., Lin, L., Wang, X., Feng, A., Yu, A., 2020. Pb(II) uptake onto nylon microplastics: interaction mechanism and adsorption performance. *J. Hazard Mater.* 386, 121960 <https://doi.org/10.1016/j.jhazmat.2019.121960>.
- Tang, S., Lin, L., Wang, X., Yu, A., Sun, X., 2021. Interfacial interactions between collected nylon microplastics and three divalent metal ions (Cu(II), Ni(II), Zn(II)) in aqueous solutions. *J. Hazard Mater.* 403, 123548 <https://doi.org/10.1016/j.jhazmat.2020.123548>.
- Tiseo, I., 2021. Global Plastics Industry - Statistics & Facts [WWW Document]. URL (accessed 3.26.21).
- Toussaint, B., Raffael, B., Angers-Loustau, A., Gilliland, D., Kestens, V., Petrillo, M., Rio-Echevarria, I.M., Van den Eede, G., 2019. Review of micro- and nanoplastic contamination in the food chain. *Food Addit. Contam. Part A Chem Anal Control Expo Risk Assess.* <https://doi.org/10.1080/19440049.2019.1583381>.
- Turner, A., Wallerstein, C., Arnold, R., Webb, D., 2019. Marine pollution from pyroplastics. *Sci. Total Environ.* 694 <https://doi.org/10.1016/j.scitotenv.2019.133610>.
- Ukuhor, H.O., 2021. The interrelationships between antimicrobial resistance, COVID-19, past, and future pandemics. *J. Infect. Publ. Health.* <https://doi.org/10.1016/j.jiph.2020.10.018>.
- Van Boeckel, T.P., Brower, C., Gilbert, M., Grenfell, B.T., Levin, S.A., Robinson, T.P., Teillant, A., Laxminarayan, R., 2015. Global Trends in Antimicrobial Use in Food Animals. In: *Proc Natl Acad Sci U S A*. <https://doi.org/10.1073/pnas.1503141112>.
- van Hengel, A.J., Marin, L., 2019. Research, innovation, and policy: an alliance combating antimicrobial resistance. *Trends Microbiol.* <https://doi.org/10.1016/j.tim.2018.12.005>.
- Viršek, M.K., Lovšin, M.N., Koren, Š., Kržan, A., Peterlin, M., 2017. Microplastics as a vector for the transport of the bacterial fish pathogen species *Aeromonas salmonicida*. *Mar. Pollut. Bull.* <https://doi.org/10.1016/j.marpolbul.2017.08.024>.
- Wang, W., Gao, H., Jin, S., Li, R., Na, G., 2019. The ecotoxicological effects of microplastics on aquatic food web, from primary producer to human: a review. *Ecotoxicol. Environ. Saf.* 173, 110–117. <https://doi.org/10.1016/j.ecoenv.2019.01.113>.
- Wang, S., Xue, N., Li, W., Zhang, D., Pan, X., Luo, Y., 2020. Selectively Enrichment of Antibiotics and ARGs by Microplastics in River, Estuary and Marine Waters. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2019.134594>.
- Wang, Y., Lu, J., Engelstädter, J., Zhang, S., Ding, P., Mao, L., Yuan, Z., Bond, P.L., Guo, J., 2020. Non-antibiotic pharmaceuticals enhance the transmission of exogenous antibiotic resistance genes through bacterial transformation. *ISME J.* <https://doi.org/10.1038/s41396-020-0679-2>.
- Wang, X., Lin, Y., Zheng, Y., Meng, F., 2022. Antibiotics in mariculture systems: a review of occurrence, environmental behavior, and ecological effects. *Environ. Pollut.* 293, 118541 <https://doi.org/10.1016/j.envpol.2021.118541>.
- WHO, 2015. WHO Library Cataloguing-in-Publication Data Global Action Plan on Antimicrobial Resistance, Microbe Magazine.
- WHO, 2020. WHO Director-General's Opening Remarks at the Media Briefing on COVID-19 - 11 March 2020 - World Health Organization. World Health Organization.
- Woodall, L., Sanchez-Vidal, A., Canals, M., Paterson, G., Coppock, R., Sleight, V., Calafat, A., Rogers, A., Narayanaswamy, B., Thompson, R., 2014. The deep sea is a major sink for microplastic debris. *R. Soc. Open Sci.* 1 <https://doi.org/10.1098/rsos.140317>.
- World Health Organization, 2020. World Health Organization Clinical Management of COVID-19: Interim Guidance 27 May 2020. World Health Organization.
- Wu, P., Huang, J., Zheng, Y., Yang, Y., Zhang, Y., He, F., Chen, H., Quan, G., Yan, J., Li, T., Gao, B., 2019. Environmental occurrences, fate, and impacts of microplastics. *Ecotoxicol. Environ. Saf.* <https://doi.org/10.1016/j.ecoenv.2019.109612>.
- Xie, M., Huang, J.L., Lin, Z., Chen, R., Tan, Q.G., 2021. Field to laboratory comparison of metal accumulation on aged microplastics in coastal waters. *Sci. Total Environ.* 797 <https://doi.org/10.1016/j.scitotenv.2021.149108>.
- Yang, Y., Liu, G., Song, W., Ye, C., Lin, H., Li, Z., Liu, W., 2019. Plastics in the marine environment are reservoirs for antibiotic and metal resistance genes. *Environ. Int.* 123, 79–86. <https://doi.org/10.1016/j.envint.2018.11.061>.
- Yao, S., Cao, H., Arp, H.P.H., Li, J., Bian, Y., Xie, Z., Cherubini, F., Jiang, X., Song, Y., 2021. The role of crystallinity and particle morphology on the sorption of dibutyl phthalate on polyethylene microplastics: implications for the behavior of phthalate plastic additives. *Environ. Pollut.* 283, 117393 <https://doi.org/10.1016/j.envpol.2021.117393>.
- Yu, M., van der Ploeg, M., Lwanga, E.H., Yang, X., Zhang, S., Ma, X., Ritsema, C.J., Geissen, V., 2019. Leaching of microplastics by preferential flow in earthworm (*Lumbricus terrestris*) burrows. *Environ. Chem.* 16, 31–40. <https://doi.org/10.1071/EN18161>.
- Yu, A., Sun, X., Tang, S., Zhang, Y., Li, M., Wang, X., 2021. Adsorption mechanism of cadmium on polystyrene microplastics containing hexabromocyclododecane. *Environ. Technol. Innovat.* 24, 102036 <https://doi.org/10.1016/j.eti.2021.102036>.
- Zalasiewicz, J., Waters, C.N., Do Sul, J.A.I., Corcoran, P.L., Barnosky, A.D., Cearreta, A., Edgeworth, M., Galuszka, A., Jeandel, C., Leinfelder, R., 2016. The geological cycle of plastics and their use as a stratigraphic indicator of the Anthropocene. *Anthropocene* 13, 4–17.
- Zhang, W., Zhang, S., Wang, J., Wang, Y., Mu, J., Wang, P., Lin, X., Ma, D., 2017. Microplastic pollution in the surface waters of the Bohai Sea, China. *Environ. Pollut.* 231, 541–548. <https://doi.org/10.1016/j.envpol.2017.08.058>.
- Zhang, Y., Lu, J., Wu, J., Wang, J., Luo, Y., 2020. Potential risks of microplastics combined with superbugs: enrichment of antibiotic resistant bacteria on the surface of microplastics in mariculture system. *Ecotoxicol. Environ. Saf.* <https://doi.org/10.1016/j.ecoenv.2019.109852>.
- Zhang, Y., Wu, D., Su, Y., Xie, B., 2021. Occurrence, influence and removal strategies of mycotoxins, antibiotics and microplastics in anaerobic digestion treating food waste and co-digestive biosolids: a critical review. *Bioresour. Technol.* 330, 124987 <https://doi.org/10.1016/j.biortech.2021.124987>.
- Zhou, Y., Liu, X., Wang, J., 2019. Characterization of microplastics and the association of heavy metals with microplastics in suburban soil of central China. *Sci. Total Environ.* 694 <https://doi.org/10.1016/j.scitotenv.2019.133798>.
- Zhu, D., Ma, J., Li, G., Rillig, M.C., Zhu, Y.G., 2021. Soil plastispheres as hotspots of antibiotic resistance genes and potential pathogens, 2021 ISME J. 16 (2 16), 521–532. <https://doi.org/10.1038/s41396-021-01103-9>.